Digitalization and Credit Markets: Evidence from eInvoicing

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Abstract

Electronic invoicing (eInvoicing) makes invoices standardized, verifiable, and hard to falsify. We document its effects on banks' information and lending decisions using administrative credit data in a difference-in-differences design that exploits a regional eInvoicing mandate. Credit reallocates toward firms already relying on invoice-based financing ("invoice firms"). Invoice firms obtain more and cheaper invoice credit, with an overall credit expansion, while non-invoice firms borrow less at higher rates. Consistent with differential changes in banks' pricing and risk-ranking behavior, dispersion in loan rates and banks' risk assessments widens for invoice firms relative to non-invoice firms, and the predictive accuracy of these assessments relatively improves (worsens) for (non-)invoice firms. Overall, eInvoicing reshapes credit allocation, with uneven effects across borrowers.

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1. Introduction

Digitalization is reshaping how firms operate, make payments, and obtain financing (Berg, Fuster, and Puri 2022; Goldfarb and Tucker 2019; Adrian and Mancini-Griffoli 2021). While prior work on digital payments emphasizes their implications for financial inclusion, efficiency, and competition, it has focused on payment execution rather than the invoicing infrastructure that underpins economic transactions.[1] This distinction matters because invoices record the counterparties, timing, and economic content of such transactions. Electronic invoicing, or eInvoicing, transforms these records into standardized, traceable, and verifiable digital documents, thereby changing the quality and usability of the recorded information. Despite its growing adoption worldwide and policy relevance, there is little evidence on whether eInvoicing affects firms' access to finance or credit markets more broadly.[2]

The premise of this study is that eInvoicing can affect credit markets by changing the information firms can credibly provide to lenders. Although eInvoicing is often introduced to reduce tax evasion and misreporting to tax authorities, its relevance for lenders need not stem from the revelation of previously hidden sales or revenues.[3] Rather, eInvoicing may matter because invoices become more standardized, easier to verify, and harder to falsify or duplicate. These features can facilitate financing arrangements directly tied to invoices, such as factoring or confirming, and may also improve credit risk assessment more broadly. By reshaping the transactional information available to lenders, eInvoicing may affect both the availability and the composition of credit to firms.

Our contribution is to document the effects of eInvoicing on credit markets by exploiting a unique policy change that mandated eInvoicing adoption at the regional level and combining it with exhaustive administrative credit data. We document substan-

1. See, among others, Parlour, Rajan, and Zhu (2022); Ghosh, Vallée, and Zeng (2026); Vives (2019); Jack and Suri (2014); Dalton et al. (2024); Ouyang (2026).
2. Notably, the European Commission considers eInvoicing as a building block for a fully digital Single Market (see eInvoicing webpage). See also International Observatory on eInvoicing and Global Interoperability Forum (2025) for more information on the global adoption of eInvoicing.
3. Firms may have incentives to conceal activity from tax authorities, and eInvoicing can increase tax compliance (Bellon et al. 2022). Nevertheless, banks need not rely on firms' legally reported income as they can infer borrowers' true income for their credit decisions (Artavanis, Morse, and Tsoutsoura 2016).

tial heterogeneity across firms, depending on their reliance on invoice-based financing arrangements prior to the eInvoicing mandate. Firms that already relied on invoice-based credit ("invoice firms") benefit from an increase in both invoice-based and overall credit, while firms that did not rely on invoice-based credit ("non-invoice firms") experience a reduction in overall credit. We provide evidence consistent with a credit-supply component, driven by changes in lenders' use of information, explaining these results. Specifically, we document how lenders have relatively more (less) information to differentiate between invoice (non-invoice) firms. Overall, our results suggest that eInvoicing primarily benefits borrowers for which better transactional records of commercial activity are more relevant to credit risk assessment, while generating negative spillovers for borrowers for which such records are less relevant.

To conduct our analysis, we exploit a unique policy change that mandated eInvoicing adoption at the regional level, combined with administrative credit data from the Bank of Spain. Our setting is the introduction of TicketBAI, a mandatory eInvoicing framework rolled out in Spain's Basque Country region from 2021 onward, while the rest of Spain was not subject to a similar mandate.[4] A key advantage of this setting is that we can observe loan-level information, loan defaults, banks' internal assessments of borrower risk, and firms' balance sheets for both the Basque Country and the neighboring, unaffected regions. This allows us to implement a difference-in-differences design to identify the effects of eInvoicing on firms' financing, credit allocation, and lenders' use of information. Specifically, we compare the evolution of credit market outcomes for firms subject to the Basque eInvoicing mandate with those of firms in neighboring regions.

TicketBAI was introduced by the provincial tax authorities of Spain's Basque Country to strengthen tax compliance and digitize firms' invoicing practices. The reform requires firms to issue digital invoices using certified software that generates a unique identification code and transmits the invoice to the provincial tax authority when it is issued. While this system digitizes, standardizes, and authenticates the information, it does not grant

4. At the time of the TicketBAI reform, no eInvoicing mandate was in place in the rest of Spain. A national framework for invoicing software, commonly known as VeriFactu, was announced in 2023 and, as of October 2026, is proposed for implementation in October 2028.

lenders automatic access to eInvoices; firms must still share them voluntarily. The rollout of TicketBAI varied across the three Basque provinces of Álava, Biscay, and Gipuzkoa and included two phases: a voluntary phase, during which firms were encouraged to adopt the system early through fiscal incentives, and a mandatory phase, during which non-compliance became subject to monetary penalties. The voluntary phase began in Gipuzkoa in January 2021 and in Álava and Biscay in January 2022, while mandatory adoption followed later and at different dates across provinces.[5] For our main analysis, we make two assumptions. First, we conservatively define the beginning of the voluntary phase as the start of treatment, given the numerous and generous fiscal incentives for early compliance.[6] Second, given the absence of firm-level compliance information, we assign treatment based on the province where the firm's headquarters is located. Firms headquartered in the neighboring provinces of Burgos, La Rioja, Cantabria, and Navarre, where no comparable eInvoicing mandate was introduced during our sample period, serve as the control group.

We combine this institutional variation with administrative data from the Bank of Spain. The CIR (Central Credit Register) provides monthly loan-level information on nearly the universe of credit exposures in Spain, including origination, type, maturity, amount, interest rate, and default status. A distinctive feature of these data is the inclusion of banks' internal estimates of borrowers' probability of default (PD).[7] We link these data to annual balance sheet and income statement information from the CBI (Central Balance Sheet Data Office), which allows us to analyze firms' credit conditions and lenders' use of information. Our sample covers January 2021 to December 2024. We exclude earlier years because the COVID-19 crisis and the associated government loan guarantee program may have distorted credit market dynamics and outcomes. We also restrict the sample to micro, small, and medium enterprises, which are less likely to borrow or do business extensively outside their home province, and exclude Gipuzkoa because it entered the

5. Mandatory adoption began in 2022 for some categories of firms in Álava and Gipuzkoa, and later became universal in December 2022 and June 2023, respectively; in Biscay, it began in January 2024 for large firms and became universal in 2026.

6. When we relax this assumption and allow for differential treatment in the voluntary and mandatory phases, we find that results are generally stronger for the mandatory phase.

7. This information is available at the bank-firm-quarter level only for banks using the Internal Ratings-Based (IRB) approach under Basel rules.

reform in early 2021, leaving no reliable pre-treatment-period data. The final sample includes 24,711 firms across the treated provinces of Álava and Biscay and the control provinces of Burgos, La Rioja, Cantabria, and Navarre. Firm-, loan-, and province-level characteristics are broadly balanced across the treated and control groups.

Our empirical strategy to identify the effects of eInvoicing adoption on firms' credit availability and composition uses a standard difference-in-differences design with granular fixed effects, comparing the evolution of credit for similar firms and lending relationships between treated and control provinces. Specifically, we estimate a specification at the bank-firm-month level that includes industry×size-tercile×month fixed effects to account for credit demand factors (à la Degryse et al. 2019), bank×month fixed effects to absorb any other relevant lender-specific variation, bank×firm fixed effects to account for non-random matching, as well as standard firm-level controls. Because our sample begins with the COVID-19 crisis and related public support programs, whose impacts may have differed across treated and control groups, and potentially correlated with credit market outcomes, we control for a wide range of COVID-related factors.[8] We also estimate firm-month specifications to assess whether results aggregate at the firm level.

Our analysis yields three main sets of results. The first is on credit quantities. We find no relevant aggregate effect of eInvoicing on firms' overall credit.[9] This aggregate result, however, masks substantial heterogeneity across firms. Invoice firms in treated provinces experience a 4.15% increase in invoice credit and a 0.64% increase in total credit relative to invoice firms in the control group. These average estimates, at the bank-firm level, also aggregate at the firm level: invoice credit increases by 2.17% and total credit rises

8. Our main specification includes: i) at the province-month level, a Google Mobility index to capture potential restrictions to economic activity and the severity of the crisis, specifically, we use the change in visitors to public transport hubs relative to a baseline; ii) at the province-industry-month level, the number of temporary furloughs that benefited from public support under the *Expediente de Regulación Temporal de Empleo* (ERTE) program to capture the potentially heterogeneous decrease in firms' operating expenses; iii) at the bank-firm-month level, the share of loans subject to government guarantees provided by the Spanish *Instituto de Crédito Oficial* (ICO), to account for potential changes in the composition of borrowing, credit terms, and default dynamics.

9. While we do observe an increase in invoice-based credit by 2.50% at the firm level, consistent with eInvoicing improving the properties of invoices relevant to lenders, this effect is quantitatively modest, as invoice-based credit accounts for around 7% of total credit for the average firm in our sample.

by 1.13%. These effects operate through both an intensive-margin effect, i.e., larger credit amounts from pre-existing invoice relationships, and an extensive-margin effect, i.e., a higher likelihood of originating new invoice-based relationships. By contrast, non-invoice firms in treated provinces experience a 0.70% decrease in total credit relative to non-invoice firms in the control group at the bank-firm level, and a 1.42% decrease at the firm level. For non-invoice firms, the decline in credit operates through the intensive margin, i.e., lower credit amounts from pre-existing relationships, rather than a significant reduction in new relationship formation. Overall, these findings indicate that eInvoicing reallocates credit toward firms for which invoice records already played a relevant role in their financing mix.

To reinforce the interpretation that eInvoicing drives these effects, we also run a battery of robustness tests. First, we distinguish between the voluntary and mandatory phases of the reform and find that effects, while already present during the voluntary phase, intensify once adoption becomes mandatory. Second, we estimate a more demanding within-province triple-difference specification that accounts for province-specific time-varying shocks and obtain similar results. Third, we address the concern that our estimates may reflect improved tax compliance rather than changes in the informational properties of invoices, and find no evidence of a differential increase in sales in treated provinces after the reform.

Our second set of results is on interest rates and links the observed credit-quantity patterns to a plausible credit-supply component. After the introduction of eInvoicing, invoice firms in treated regions face a relative decline in the cost of invoice credit, while non-invoice credit becomes relatively more expensive; non-invoice firms, in turn, face higher borrowing costs. Throughout our analysis, we define the cost of credit as the average of loan interest rates weighted by the corresponding credit amount. For invoice firms, the cost of invoice credit in our bank-firm (firm) specifications falls by about 2.4% (2.1%), whereas the cost of non-invoice credit rises by about 3.9% (10.6%). For non-invoice firms, total borrowing costs increase by about 3.9% (9.6%). Taken together with the quantity results, these patterns provide evidence consistent with a supply-side component in the reallocation of credit toward invoice-based lending for invoice firms.

Our third and final set of results provides evidence consistent with changes in banks'

use of information. Specifically, we explore two complementary dimensions. First, we examine whether eInvoicing increases the cross-sectional dispersion of loan rates and banks' internal risk assessments, on the grounds that more standardized, verifiable information in digital invoices helps banks better discriminate among borrowers. Empirically, we estimate whether the reform impacted the standard deviation and the interquartile range of loan rates and banks' internal PDs, using a province-quarter-level specification with province and quarter fixed effects and a comprehensive set of time-varying province controls. We find that eInvoicing increases the dispersion of the cost of credit by roughly 17% and of internal risk assessments by roughly 14% for invoice firms, whereas dispersion does not increase for non-invoice firms. The gap between the two groups is sizable and statistically robust, suggesting that lenders differentiate more precisely among those borrowers whose information environment improves with the reform.

Second, we assess whether eInvoicing changes the predictive accuracy of banks' internal risk assessments. Specifically, we analyze changes in the predictive accuracy of internal PDs using the Area Under the Receiver Operating Characteristic curve (AUROC), which measures how well PDs rank borrowers with respect to ex-post default. We adopt a graphical difference-in-differences approach (as in Howes and Weitzner 2026), and find a relative improvement (decrease) in predictive accuracy for invoice (non-invoice) firms.[10] For invoice firms, the treated group has a pre-reform AUROC about 0.037 lower than the control group, a difference statistically significant at conventional levels. However, this gap shrinks after the reform and becomes statistically indistinguishable from zero, suggesting a substantial relative improvement in predictive accuracy. For non-invoice firms, treated and control groups display statistically similar AUROCs in the pre-reform period, whereas after eInvoicing, the AUROC for the control group is roughly 0.052 larger than for the treated group.[11]

10. We conduct extensive robustness tests to address two concerns: first, since the design does not allow us to include covariates or fixed effects as in a regression framework, AUROC comparisons may be affected by compositional differences; and second, the post-reform period overlaps with public policies introduced during the COVID-19 pandemic. These tests include matching on observable characteristics, restricting the sample to stayers, splitting by relationship cohorts, and excluding observations in which all outstanding loans are either ICO-guaranteed or under payment moratoria.

11. These differences are also economically meaningful, as even a 0.01 improvement in AUROCs *"is considered a noteworthy gain in the credit scoring industry"* (Iyer et al. 2016, p. 1565).

Overall, these results are consistent with eInvoicing affecting banks' pricing and the quality of risk assessments, particularly for firms whose financing is more tightly linked to invoices. This suggests that banks reallocate information production and screening efforts toward those firms, and away from borrowers whose invoice records are less informative.

In conclusion, our findings suggest that eInvoicing reshapes credit allocation by changing the information available to lenders, with uneven consequences across borrowers. When new digital records are more informative for some borrowers than for others, lenders' improved ability to discriminate creates winners and losers, tightening credit precisely for those borrowers whose activity is least represented digitally. From a policy perspective, this suggests evaluating invoice digitalization initiatives not only by their average effects but also by their distributional consequences to mitigate unintended outcomes.

1.1 Related Literature

We contribute to four streams of literature. The first one studies digitalization in finance, with a particular focus on digital payments and the broader transformation of financial intermediation (Vives 2019). Empirical work shows that digital payments can expand financial inclusion and improve welfare (Ouyang 2026; Jack and Suri 2014; Suri and Jack 2016; Dalton et al. 2024), support household and business activity (Dubey and Purnanandam 2023), and strengthen monetary policy transmission and competition for deposits (Huang 2024; Liang, Sampaio, and Sarkisyan 2024; Sarkisyan 2023). Related evidence shows that cash shortages can reduce economic activity and accelerate the adoption of alternative payment technologies (Chodorow-Reich et al. 2020). From a theoretical point of view, Parlour, Rajan, and Zhu (2022) argue that FinTech competition in payment services changes banks' information and lending decisions, with ambiguous welfare effects. Our contribution is to shift the focus from the digital execution of payments to an equally fundamental input in corporate finance: the invoicing infrastructure that underpins economic transactions. We show that digitizing invoices matters for firms' financing, highlighting the potential spillovers of digitalization policies to credit markets.

A second line of research studies how financial technology can shape credit access and

market structure by changing the information set available to lenders (Berg, Fuster, and Puri 2022). Theoretical work argues that borrower-controlled data sharing can improve competition and screening but may also strengthen the market power of better-informed lenders (He, Huang, and Zhou 2023). Empirical work shows that open banking and FinTech can broaden credit access, facilitate new lending relationships, and alter credit supply and market structure (Babina et al. 2025; Hau et al. 2024; Erel and Liebersohn 2022; Fuster et al. 2019). Relatedly, a growing number of studies show that digital footprints and transaction data can improve credit-risk assessment and expand access to credit, although the gains depend on the precision of the information and lenders' capacity to process it (Berg et al. 2020; Ghosh, Vallée, and Zeng 2026; Alok et al. 2024; Huang et al. 2023; Di Maggio and Ratnadiwakara 2026; Russel, Shi, and Clarke 2024). We analyze a distinct source of digital information: standardized, verifiable invoice records generated by eInvoicing. By documenting heterogeneous effects across firms, our evidence highlights that the gains from digitalization in credit markets are not uniform and speaks to the broader question of who benefits from digitalization.

The third literature stream concerns the role of information in credit markets. A central question in this literature is whether greater quality and availability of public information substitutes for or complements banks' private information. On the one hand, better public signals may reduce banks' incentives to invest in costly screening and monitoring by compressing informational rents (Hauswald and Marquez 2003; Dell'Ariccia and Marquez 2004; Vives and Ye 2025a,b). On the other hand, they may complement banks' monitoring and screening efforts by making private information more valuable or easier to use (Pagano and Jappelli 1993; Karapetyan and Stacescu 2014; Padilla and Pagano 1997; Bennardo, Pagano, and Piccolo 2015). Empirical work, mostly focused on information-sharing arrangements among lenders, offers mixed evidence (Brown, Jappelli, and Pagano 2009; Doblas-Madrid and Minetti 2013; Jappelli and Pagano 2002; Sutherland 2018; Hertzberg, Liberti, and Paravisini 2011; Cheng and Degryse 2010). Our contribution is to bring this question to a new setting in which firms generate authenticated, verifiable, and standardized transaction data through eInvoicing. We document how eInvoicing affects banks' use of information and

that the consequences of improved hard information are not uniform, but depend on how relevant the new records are to lenders' screening and monitoring of different borrowers.

Finally, our paper contributes to the nascent literature on eInvoicing. Virtually all the existing studies focus on the implications of digitizing invoicing for tax compliance. The empirical evidence points to the positive role of eInvoicing adoption in value-added tax (VAT) compliance and collection, both in developing countries (Bellon et al. 2022; Bellon, Dabla-Norris, and Khalid 2023; Kotsogiannis et al. 2025; Mascagni, Mengistu, and Woldeyes 2021; Ali et al. 2021; Hakizimana and Santoro 2023) and in more advanced economies (Tao and Li 2026; Fan et al. 2018; Heinemann and Stiller 2025). Relative to these studies, we are the first to document that the effects of eInvoicing adoption extend beyond tax compliance and can spill over into firms' financing and credit markets.

2. The Basque eInvoicing Mandate

TicketBAI is a regulatory initiative introduced by the provincial tax authorities of Spain's Basque Country, a region composed of the three provinces Álava, Biscay, and Gipuzkoa (see Figure IA.I), to digitalize the documentation of business activity.[12, 13] The reform mandates that firms issue invoices using certified invoicing software that automatically transmits structured, tamper-proof invoice data to tax authorities in real time. Each invoice contains a cryptographic hash and QR code, ensuring authenticity and traceability.

The reform thus improves the credibility and verifiability of firms' commercial activity by standardizing transaction-level information, as all firms have to use such software when

12. TicketBAI is part of a broader process of invoice digitalization in Spain. As of October 2026, the Spanish tax authority launched separate nationwide requirements for invoicing software (VeriFactu) and business-to-business eInvoicing. Implementation of the former is proposed for October 2028, while the latter will become mandatory in October 2027 for businesses with annual turnover exceeding €8 million, and October 2028 for the remainder. In both cases, after our sample period ends. Link: https://sede.agenciatributaria.gob.es/Sede/iva/sistemas-informaticos-facturacion-verifactu.html.

13. While not a digitalization initiative per se, some firms in Spain are subject to the *Suministro Inmediato de Información* (SII), a VAT compliance and tax-control system for large VAT taxpayers introduced in 2017. SII mandates that specific firms, e.g., those with annual turnover above €6 million, *self-report* VAT-book billing records to the tax authority within four days. Link: https://sede.agenciatributaria.gob.es/Sede/en_gb/iva/suministro-inmediato-informacion/informacion-general.html. Notably, our findings are unlikely to be driven by the SII. See Internet Appendix C.6 for a detailed discussion.

invoicing. Firms can then provide this standardized, verifiable documentation of accounts receivable and contractual relationships to banks, thereby changing lenders' information set. Because firms must still voluntarily share electronic invoices and authorize lenders to access them, our estimates likely represent a lower bound on the effects of reforms or technologies that would give lenders automatic access to these data, provided that such access does not generate significant general equilibrium effects.

The possible relevance of such a mandate for credit stems from anecdotal evidence suggesting that banks actively rely on invoice-related information when assessing firms' credit risk. Industry and policy discussions also point in the same direction, highlighting that the standardized format and near-real-time generation of eInvoicing data can facilitate the assessment of ceded invoices and reduce frictions in receivables financing.[14] This view is also consistent with recent market initiatives aimed at improving the reliability of invoice-based financing. For example, the interbank platform *InBlock*, launched after our sample period (October 2025), was designed to reduce the risk of financing non-existent or duplicated invoices, indicating that lenders value invoice-level information that is easier to trust and process.[15]

Several features of TicketBAI's introduction are particularly relevant for our empirical design. First, adoption was limited to the Basque Country – it did not occur in other regions of Spain – and unfolded gradually across provinces. It took place in two phases: a voluntary phase supported by substantial fiscal incentives and a mandatory phase enforced through binding legal penalties. Voluntary adoption began in Gipuzkoa in January 2021, and in Álava and Biscay in January 2022. The start of mandatory adoption varied by province and firm type. In Álava, eInvoicing was compulsory for tax professionals starting in April 2022, followed by a sectoral rollout that culminated in December 2022, when every firm had to comply with the eInvoicing mandate. In Gipuzkoa, mandatory adoption started in July 2022 and rolled out across sectors through June 2023. In Biscay, mandatory adoption

14. See, for instance, the European Commission's stance on eInvoicing. Link: https://ec.europa.eu/digital-building-blocks/sites/spaces/DIGITAL/pages/467108637/eInvoicing.
15. See *Asociación Española de Factoring*'s newsletters about *InBlock* and *Factura Electrónica* for additional information. Link: https://factoringasociacion.com/publicaciones-y-newsletter/.

began in January 2024 for large firms and was phased in for SMEs through January 2026. Figure IA.II summarizes the different phases of the rollout across the three provinces.

Second, we do not directly observe firm-level compliance. Our design identifies the effect of province-time-level exposure to TicketBAI rather than actual firm compliance. In the absence of firm-level adoption data, we rely on the timing of province-wide implementation in our identification strategy.

Third, the institutional design included both strong incentives for early compliance and stringent penalties for non-compliance once the mandatory phase began. On the one hand, Álava and Gipuzkoa allowed firms to deduct 30–60% of eligible implementation costs from taxable income, while Biscay offered a 30% deduction, capped at €5,000.[16] These incentives likely induced meaningful adoption even before legal mandates took effect. On the other hand, firms could be fined €2,000 per non-compliant invoice, with systematic violations penalized up to 20–30% of prior-year turnover. Severe infractions involving data manipulation carried a minimum fine of €40,000.[17] Taken together, these incentives and penalties make it plausible that province-time-level exposure is related to actual adoption already during the voluntary period, even though we do not observe firm-level compliance directly. For this reason, we conservatively assume the treatment begins with the voluntary phase and continues through the mandatory phase.

Fourth, although the legal rollout of mandatory adoption also differed across industries, we cannot exploit this variation in our identification strategy for several reasons. On the one hand, TicketBAI uses the *Impuesto sobre Actividades Económicas* (Economic Activity Tax – IAE) classification to determine when firms become subject to mandatory adoption, whereas the industry information in our data is based on the standard NACE classification. Because there is no clean one-to-one mapping between the two systems, industry-level treatment assignment within a province would be measured with substantial error.[18] On the other hand, industry-specific mandatory deadlines are an imperfect proxy

16. See Biscay's Norma Foral 8/2023 and accompanying Batuz decrees; Álava's Normative Decree of Fiscal Urgency DN 9/2022; Gipuzkoa's Foral Decree DF 32/2020.
17. These figures refer to Biscay; see https://www.bizkaia.eus/es/web/comunicacion/noticias/-/news/detailView/20665. Álava and Gipuzkoa have similar rules.
18. For example, NACE codes 6201, 6202, and 6209 map into both IAE sectors 2-76 and 1-845, with

for treatment timing, since firms could and had incentives to adopt earlier during the voluntary phase. Finally, timing differences across industry blocks are often short, which limits the relevant variation available for identification. For these reasons, we do not exploit the industry-specific rollout and instead rely on treatment timing defined at the province-time level, starting with the voluntary period.

3. Data

In this section, we describe the data sources used in the analysis, the sample construction, and the main sample characteristics. We also present covariate balance tests between treated and control provinces and descriptive evidence on invoice-based credit use.

3.1 Data Sources and Sample Construction

Our analysis combines two administrative data sources provided by the Bank of Spain. The Central Credit Register (CIR) provides monthly loan-level information on nearly the universe of credit exposures in Spain, as its reporting threshold is set at €3,000. It includes detailed information on loan terms and outcomes, such as origination, type, maturity, amount, interest rate, and default status. For banks using the Internal Ratings-Based (IRB) approach under Basel regulation, the CIR also reports internal estimates of borrowers' one-year probability of default (PD), which we use to analyze lenders' use of information. We complement these data with annual firm-level balance sheet and income statement information from the Central Balance Sheet Data Office (CBI), including total assets, sales, leverage, liquidity, tangibility, profitability, age, and the firm's location and industry.[19] Merging CIR and CBI allows us to construct a high-frequency loan-firm panel with matched balance-sheet, credit, and ex-post default information. To account for the COVID-19 crisis and related policy responses, we also collect province-industry-time

the former subject to mandatory implementation in Álava starting in July 2022 and the latter in December 2022. This type of mismatch arises because the IAE classification distinguishes among business, professional, and artistic activities for tax purposes, whereas NACE classifies firms by their principal economic activity.

19. Firm size categories follow the European Commission Recommendation 2003/361/EC.

information on temporary furloughs under the *Expediente de Regulación Temporal de Empleo* (ERTE) program and community mobility indices at the province-time level from Google COVID-19 Mobility Reports.

We obtain loans issued to firms headquartered in the three provinces of the Basque Country – Álava, Biscay, and Gipuzkoa – and the four neighboring provinces – Burgos, Cantabria, La Rioja, and Navarre. Because treatment is assigned at the province level based on the location of a firm's headquarters, we restrict the sample to micro enterprises and small and medium-sized enterprises (SMEs), which are less likely to borrow or operate extensively outside their headquarters' province.[20] For our main analysis, we further restrict the sample from January 2021 to December 2024 and to firms active in 2021.[21] We exclude earlier years because the COVID-19 crisis and the associated loan-guarantee and support programs may have distorted credit conditions and outcomes. This choice also implies excluding Gipuzkoa, which entered the voluntary phase of the reform in early 2021 and therefore lacks a reliable pre-treatment period.[22] Our treated provinces are thus Álava and Biscay, while Burgos, Cantabria, La Rioja, and Navarre serve as controls. Throughout the analysis, we aggregate loan-level information to the bank-firm-month level unless otherwise specified. We winsorize all variables at the 1st and 99th percentiles.

Our final dataset includes 24,711 firms, accounting for almost 1.9 million bank-firm-time observations across treated and control provinces. In the next subsections, we report the balance tables between treated and control provinces (see Table 1), and discuss the differences between invoice and non-invoice firms (see Table IA.II). Additional summary statistics are reported in Internet Appendix B.

20. We also remove firms whose monthly credit exceeds €20 million.
21. In addition, we drop all observations of firms for which, in any month, log total credit deviates from the firm's own average over January 2021–December 2024 by an amount below the 2.5th or above the 97.5th percentile of the pooled distribution of such deviations.
22. Nevertheless, the inclusion of Gipuzkoa does not meaningfully change the results. See Internet Appendix C.4.

3.2 Sample Characteristics

Table 1 reports summary statistics and a covariate balance analysis between treated and control provinces. Panel A describes bank-firm-level characteristics for the full sample of SME bank-firm relationships as of the end of 2021. Treated and control bank-firm relationships look similar along most dimensions. The average bank-firm relationship involves about €208,000 in total credit in treated provinces and €223,000 in control provinces, with comparable risk profiles: the average probability of default is 5.36% in treated versus 5.63% in control provinces, and the realized default rate is 2.44% versus 2.89%. Pricing and maturity are also closely aligned, with average interest rates of 3.06% in treated provinces and 2.86% in control provinces, and an average maturity of about 80 months in both groups. Invoice-based credit accounts for a similar share of total credit in the two groups, around 6%. None of these differences is statistically significant. The only meaningful difference is in the share of ICO-guaranteed credit, which is 28% in treated versus 34% in control provinces.

Panel B reports firm-level characteristics and again indicates that the two groups are broadly comparable. Firms in treated and control provinces have a similar number of loans (approximately 7) and bank relationships (2.2). The share of firms classified as invoice firms (see next subsection) is almost identical across the two groups, at 23% in treated versus 24% in control provinces. Treated and control firms are also of comparable size, holding about €1.6 million in assets and generating about €1.4 million in annual sales, and they look alike in terms of leverage (0.30 vs. 0.33), liquidity (0.66 vs. 0.62), profitability (0.02 vs. 0.03), age (about 17 years), and do not differ significantly in exposure to ERTE. The only statistically significant difference is in tangibility, which is somewhat lower in treated than in control provinces (0.23 vs. 0.29).

Panel C reports province-level aggregates and confirms the same pattern of broad similarity. Treated and control provinces have comparable IRB banks' market shares (0.46 vs. 0.52), and industry composition, with shares of credit to agriculture, mining, manufacturing, construction, retail/wholesale, transport, and hospitality that are statistically indistinguishable across the two groups. Mobility patterns as measured by the

Google Mobility Index are also statistically indistinguishable. The main difference is that treated provinces have a slightly lower share of SMEs (0.93 vs. 0.96). Overall, Table 1 suggests that the Basque Country's neighboring provinces are a plausible control group, since treated and control areas are broadly comparable on observables relevant to credit demand, credit supply, and lenders' risk assessments.

3.3 Stylized Facts about Invoice-Based Credit Use

The TicketBAI reform affects the properties of invoices, and therefore, its most likely effect is to change the information environment surrounding invoice-based credit. We define invoice-based credit as the credit granted based on the purchase of invoices, bills, or other collection rights arising from the deferral of payments associated with the sale of goods or the provision of services.[23] Because the reform standardizes, authenticates, and makes the transaction-level information underlying these instruments more verifiable, its effects are likely strongest for firms for which invoice-based financing is already an important part of their financing mix before the reform.

The data show high persistence in invoice-based credit use. Firms that use invoice-based credit in a given quarter continue to do so in the next quarter with probability 90.2%, whereas firms that do not use it remain non-users with probability 97.9%. At the yearly horizon, the corresponding transition probabilities are 86.3% and 97.1%, respectively (see Table 2). Transitions between the two states are therefore rare, suggesting that invoice-based credit is a stable feature of firms' financing structure rather than a transitory borrowing choice.

Accordingly, we classify firms by their pre-reform use of invoice-based credit. Specifically, we define an "invoice firm" as a firm with at least one outstanding invoice-based loan in the pre-reform period in our sample—that is, in 2021—and a "non-invoice firm" otherwise. Invoice and non-invoice firms differ markedly along several dimensions (see Table IA.II). Invoice firms account for 23.3% of firms in the sample, but they are substantially larger, older, and more connected to the banking system than non-invoice firms. They also differ

23. See Internet Appendix B.1 for the precise set of Bank of Spain categories included in our definition.

Table 1. Descriptive Statistics: Treated vs. Control Groups

This table presents descriptive statistics for the treated and control groups. All variables are measured as of the end of 2021, the pre-reform period, and are defined in Table IA.I. Columns (1) and (2) report the mean for the treated and control groups, respectively. Column (3) reports the difference between the treated and control groups, and Column (4) reports the p-value of a two-sided test for the equality of means across the treated and control groups. Standard deviations appear in parentheses, except for N, for which the relative share appears in parentheses. Panel A presents bank-firm-level variables for the full sample of bank-firm relationships as of the end of 2021. Panel B describes firm-level variables. Panel C reports province-level aggregates. For the tests of equality of means, standard errors are clustered at the province×industry level in Panels A and B; Panel C reports homoskedastic standard errors. p-values are reported to two decimal digits, with values below 0.01 reported as < 0.01. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Treated	Control	Diff.	P-value
	(1)	(2)	(3)	(4)
Panel A: Bank-Firm Characteristics				
N	20,309 (43.90%)	25,950 (56.10%)		
Total credit, thousand €	207.82 (567.49)	222.51 (534.38)	–14.69	0.53
Probability of default, %	5.36 (16.79)	5.63 (17.89)	–0.27	0.61
Default rate, %	2.44 (15.43)	2.89 (16.75)	–0.45	0.17
Interest rate, %	3.06 (3.32)	2.86 (2.65)	0.20	0.12
Maturity, months	80.31 (51.14)	79.59 (53.50)	0.72	0.89
Share of invoice-based credit	0.06 (0.20)	0.07 (0.20)	0.00	0.90
Share of ICO-guaranteed credit	0.28 (0.39)	0.34 (0.41)	–0.05	$< 0.01^{***}$
Panel B: Firm Characteristics				
N	9,797 (44.05%)	12,442 (55.95%)		
N. loans	6.58 (12.66)	6.82 (11.21)	–0.24	0.74
N. bank relationships	2.20 (1.78)	2.28 (1.83)	–0.08	0.57
N. IRB bank relationships	0.90 (0.92)	0.96 (0.90)	–0.06	0.36
Invoice firm	0.23 (0.42)	0.24 (0.42)	–0.01	0.89
Total assets, million €	1.60 (3.60)	1.63 (3.53)	–0.02	0.92
Sales, million €	1.40 (3.39)	1.44 (3.49)	–0.05	0.86
Leverage	0.30 (0.30)	0.33 (0.29)	–0.03	0.13
Liquidity	0.66 (0.31)	0.62 (0.29)	0.04	0.35
Tangibility	0.23 (0.27)	0.29 (0.27)	–0.06	0.04^{**}
Profitability	0.02 (0.19)	0.03 (0.17)	–0.01	0.25
Age	17.01 (10.93)	17.63 (11.54)	–0.62	0.47
ERTE	81.86 (152.21)	32.17 (61.26)	49.70	0.18
Panel C: Province Characteristics				
N	2 (33.33%)	4 (66.67%)		
Share of SMEs	0.93 (0.00)	0.96 (0.01)	–0.03	0.03^{**}
IRB banks' market share	0.46 (0.07)	0.52 (0.05)	–0.06	0.27
Corporate credit market HHI	0.10 (0.00)	0.14 (0.02)	–0.04	0.10
Share of credit to :				
Agriculture	0.02 (0.01)	0.03 (0.01)	–0.01	0.18
Mining	0.00 (0.00)	0.00 (0.00)	0.00	0.71
Manufacturing	0.30 (0.09)	0.32 (0.08)	–0.02	0.86
Construction	0.14 (0.03)	0.12 (0.03)	0.02	0.44
Retail/Wholesale	0.23 (0.03)	0.25 (0.04)	–0.02	0.60
Transport	0.04 (0.02)	0.07 (0.03)	–0.03	0.25
Hospitality	0.05 (0.00)	0.06 (0.02)	–0.01	0.44
Google Mobility Index	–16.68 (4.84)	–1.99 (19.32)	–14.69	0.37

Table 2. Persistence in Invoice-Based Credit Use.

This table reports transition probabilities in firms' use of invoice-based credit. Rows indicate whether firm f has positive invoice-based credit in period t, while columns indicate whether the same firm has positive invoice-based credit in period $t+1$. Firms classified as having no invoice-based credit may still use other forms of bank credit. Panel A reports transitions at the firm-quarter level; Panel B reports transitions at the firm-year level.

Panel A: Firm-quarter level		
	No Invoice-based Credit$_{f,t+1}$	Invoice-based Credit$_{f,t+1}$
No Invoice-based Credit$_{f,t}$	97.9%	2.1%
Invoice-based Credit$_{f,t}$	9.8%	90.2%
Panel B: Firm-year level		
	No Invoice-based Credit$_{f,t+1}$	Invoice-based Credit$_{f,t+1}$
No Invoice-based Credit$_{f,t}$	97.1%	2.9%
Invoice-based Credit$_{f,t}$	13.7%	86.3%

systematically along other margins: invoice firms are more leveraged, hold a larger share of liquid assets, and rely more heavily on ICO-guaranteed credit, while exhibiting lower tangibility and slightly lower return on assets. These patterns are consistent with business models in which trade receivables are a recurrent and important component of working capital, and in which physical collateral plays a relatively smaller role. Taken together, the persistence in invoice-based credit use and the sharp differences in firm characteristics suggest that invoice and non-invoice firms rely on structurally different financing strategies rather than representing temporary users and non-users of the same credit product.

This motivates analyzing the effects of eInvoicing separately for invoice and non-invoice firms. A natural question is whether treated and control firms are comparable within each group. Table IA.III reports a covariate balance test for invoice firms and non-invoice firms across treated and control provinces. Overall, within each group, treated and control firms are balanced on most observable characteristics. Invoice firms in treated provinces exhibit higher liquidity, lower tangibility, lower profitability, and a lower share of ICO-guaranteed loans than invoice firms in control provinces. Non-invoice firms in treated and control provinces are even more similar, with differences only in tangibility and the share of

ICO-guaranteed loans.

4. eInvoicing and Credit Market Dynamics

After describing basic facts about firms, credit markets, and invoice-based credit use in our sample, we analyze the effects of eInvoicing on firms' credit. First, we examine whether eInvoicing adoption increases the use of invoice-based credit (hereafter, invoice credit). By requiring firms to issue standardized, digitally signed invoices, the reform alters the informational properties of the records that underlie business transactions, as discussed in Section 2. This should be particularly relevant for credit instruments backed by invoices, where lenders rely on information about the existence, authenticity, and quality of the underlying commercial claims. In this sense, eInvoicing lowers lenders' information-processing costs: it reduces both the cost of verifying the assets underlying invoice-based lending and the cost of standardizing invoice information before incorporating it into screening, monitoring, and credit-allocation decisions. This mechanism is closely related to theories in which more verifiable cash flows and pledgeable claims relax financing constraints and support credit provision (Hart and Moore 1994; Holmström and Tirole 1997). We therefore expect eInvoicing to increase firms' use of invoice credit.

We next analyze whether this increase in invoice credit reflects substitution across financing instruments or broader credit growth. If eInvoicing mainly improves the relative attractiveness of invoice-backed lending, firms may shift borrowing toward invoice-based credit without increasing total credit, leading to substitution. By contrast, if standardized and reliable invoice records also improve lenders' assessment of borrower quality or reduce monitoring costs more broadly, total credit may increase as well.[24] This view is consistent with the broader idea that more reliable hard information can improve credit decisions (Liberti and Petersen 2019), and with more recent evidence that digital records and alternative data can facilitate borrower screening (Berg et al. 2020; Di Maggio and

24. The effect on total credit is, in principle, ambiguous. Better information may increase or reduce overall credit depending on whether, on average, the information revealed by eInvoicing leads lenders to reassess firms more positively or more negatively.

Ratnadiwakara 2026).

Relatedly, we analyze whether the effect of eInvoicing is heterogeneous across firms. We argue that the reform may have larger effects for firms whose financing is already closely linked to invoices and receivables, both because invoice records are likely to be more informative for lenders' credit decisions and because the reform may lower the cost of expanding an existing financing technology. For firms with no prior use of invoice-based credit, the effect might not be so clear-cut. While they may also benefit if the reform reduces adoption frictions and facilitates entry into this form of financing, the effect could also be negative if lenders face capacity constraints in processing additional information (Campbell, Loumioti, and Wittenberg-Moerman 2019; Mariathasan and Zhuk 2022) or if they reallocate credit from one type of firm to the other due to better information about invoice firms. In such a case, the reform may reallocate screening effort and credit supply toward firms for which invoice data are most informative and away from firms for which such data are less useful. More generally, to the extent that invoice data improve lenders' ability to assess firms' commercial activity, the effects on total credit should be stronger among firms for which invoices are more informative about credit risk. Which force dominates, and for which firms, is ultimately an empirical question.

4.1 Identification Strategy

To test these hypotheses, we employ a standard difference-in-differences (DiD) design that exploits the introduction of the TicketBAI eInvoicing reform in the Basque Country as a source of plausibly exogenous variation in firms' eInvoicing use. Our design rests on three features. First, Álava and Biscay share the same start of the voluntary adoption phase, in January 2022, even though the timing of mandatory adoption later differs across the two provinces. Second, we define treatment as starting with the voluntary phase rather than the mandatory one. As discussed in Section 2, both provinces offered substantial incentives for early compliance, making it likely that some firms adjusted their invoicing systems before adoption became legally mandatory. We therefore use January 2022 as the common treatment start date for Álava and Biscay, and regard this choice as conservative,

since any resulting misclassification would attenuate our estimates toward zero.[25] Third, because our sample begins in 2021 for the reasons discussed in Section 3, this treatment definition leaves no reliable pre-reform period for Gipuzkoa, so we exclude that province from the analysis.[26]

Formally, we estimate the following specification from January 2021 to December 2024 at the bank-firm-month level:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}. \quad (1)$$

where f, b, p, i, s, and t index firm, bank, province, industry, size tercile, and month, respectively. The dependent variable $Y_{f,b,t}$ denotes either the amount of total credit, invoice credit, or non-invoice credit extended by bank b to firm f in month t. Because invoice-based credit is inherently short-term, the data contain many bank-firm-month observations in which this outcome equals zero. We therefore estimate Equation (1) using Poisson Pseudo-Maximum Likelihood (PPML), which allows us to retain observations with value zero and to interpret the coefficients as semi-elasticities (Chen and Roth 2024).[27] Standard errors are clustered at the province×month level.

Our coefficient of interest is β, which captures how credit outcomes evolve after January 2022 for firms in treated provinces relative to firms in control provinces. The identifying variation comes from comparing the same bank-firm relationships before and after the reform across treated and control areas. The granular set of fixed effects in Equation (1) helps make this comparison as informative as possible. First, we include bank×month fixed effects ($\alpha_{b,t}$), which absorb all lender-specific shocks that vary over time. These fixed effects matter in our setting because banks may adjust credit supply in response to macroeconomic conditions, monetary policy, or internal strategies unrelated to TicketBAI. Second, we include industry×size-tercile×month fixed effects ($\alpha_{i,s,t}$), which

25. We also repeat our analysis employing a staggered DiD design in which we explicitly distinguish between the voluntary and mandatory periods. See Section 4.5 for more details.
26. In Internet Appendix C.4, we show that including Gipuzkoa in our baseline analysis does not meaningfully change our main results.
27. Specifically, we use the `ppmlhdfe` routine in Stata developed by Correia, Guimarães, and Zylkin (2020).

absorb time-varying credit demand factors common to firms of similar size operating in the same industry (Degryse et al. 2019). This lets us compare treated and control firms facing similar sectoral and size-specific credit-demand conditions. Third, we include bank×firm fixed effects ($\alpha_{f,b}$), which account for all time-invariant features of a given lending relationship, including non-random matching between banks and borrowers.

Our identifying assumption is that, after absorbing lender-specific shocks, time-varying factors common to firms in the same industry and size class, and time-invariant features of bank-firm relationships, treated and control firms would have followed similar trends in credit outcomes absent the reform. In other words, conditional on our fixed effects, the introduction of TicketBAI is unrelated to other factors that differentially affect the evolution of credit in treated provinces after January 2022. As discussed in Section 3, the pre-reform covariate balance between treated and control firms supports the plausibility of this assumption (see Table 1).

To further reduce potential bias, we control for a rich set of time-varying firm and bank-firm controls ($X_{f,t-1}$), as well as factors related to the COVID-19 crisis and its policy response. At the firm level, we include measures of size, leverage, liquidity, tangibility, profitability, and age. At the bank-firm level, we include the lagged share of firm f's outstanding credit with bank b covered by ICO guarantees, which captures differences in the composition of borrowing and in banks' effective risk exposure, as well as an indicator for whether the firm is in default on any outstanding credit position with bank b. At the province-industry-month level, we include the number of workers covered by the *Expediente de Regulación Temporal de Empleo* (ERTE) program, which captures sector-specific labor-market support during the pandemic. We also control for a Google Mobility index that tracks changes in visitors to public transport hubs relative to a baseline (from Google COVID-19 Mobility Reports) at the province-month level, to account for heterogeneity in local restrictions to economic activity and in the severity of the pandemic shock. Overall, these controls reduce concerns that differences in firm composition or exposure to COVID-related disruptions drive our estimates.

4.2 Main Results: Full Sample

We begin by examining whether eInvoicing affects firms' credit use in the full sample. Table 3, Panel A, reports the results from estimating Equation (1) using invoice-based credit, non-invoice-based credit, and total credit as outcomes at the bank-firm level.

The estimates show an increase in invoice credit for treated firms following the introduction of TicketBAI relative to the control group. In column (1), the coefficient on *Post* × *Treated* is 0.0462 and statistically significant at the 1% level. Under our PPML specification, this implies an increase of about $100 \times (\exp 0.0462 - 1) = 4.73\%$ in invoice-based credit.[28] By contrast, columns (2) and (3) show essentially no effect on non-invoice credit or total credit: the estimated effects are -0.04% and 0.03%, respectively, and both are statistically insignificant. Thus, at the bank-firm level, eInvoicing increases the use of invoice-based credit relative to the control group, but without affecting either total borrowing or other forms of credit.[29]

These results are consistent with the idea that eInvoicing makes invoice-based lending more attractive. By improving the informational properties of invoices, the reform appears to strengthen the use of credit instruments based on invoices and receivables. At the same time, the absence of any effect on total credit at the bank-firm level suggests that the increase in invoice credit relative to the control group does not mechanically translate into a broader increase in borrowing within individual lending relationships.

Next, we examine whether these relationship-level effects aggregate to the firm level or are offset by substitution effects across banks. Because our previous estimates capture the average effect across lending relationships, a positive effect at the bank-firm level need not imply higher firm-level borrowing if firms expand invoice credit with some lenders while reducing it with others. To assess whether the reform changes firms' overall borrowing rather than only the composition of credit across lending relationships, we re-estimate the

28. Throughout this section, we report coefficient estimates as percentage effects, computed as $100 \times \left(\exp \hat{\beta} - 1\right)$.

29. This is likely because invoice-based credit accounts for around 7% of total credit for the average firm in our sample.

Table 3. eInvoicing and Credit Supply

This table reports estimates of β from the following regressions over our main sample from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. In Panel A, the unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

In Panel B, the unit of observation is a firm-month, credit outcomes are aggregated across all banks lending to firm f in month t, and we estimate:

$$\mathbb{E}[Y_{f,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_f + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_{\bar{b}} \cdot \bar{X}_{b,f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

In Panel A, $Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. In Panel B, $Y_{f,t}$ is the corresponding firm-level amount. The dependent variable is invoice-based credit in column 1, non-invoice credit in column 2, and total credit in column 3. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. Panel A includes industry×size-tercile×month, firm×bank, and bank×month fixed effects. Panel B includes industry×size-tercile×month and firm fixed effects. In both panels, $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, and age. In Panel A, this vector also includes bank-firm-level controls: the share of bank-firm credit guaranteed by ICO and an indicator for whether the firm is in default with that bank. In Panel B, these two variables are replaced by their firm-level counterparts: the share of the firm's total credit guaranteed by ICO and an indicator for whether the firm is in default with any bank. Average bank controls in Panel B are denoted by $\bar{X}_{b,f,t-1}$ and are computed as firm-level weighted averages, with weights based on each bank's share of the firm's credit. These controls include bank profitability, liquidity, size, non-performing loan ratio, and capitalization. In both panels, $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. The row *Controls* indicates that all specifications include firm and Covid controls in Panel A, and firm, Covid, and average bank controls in Panel B. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice	Non-Invoice	Total
	(1)	(2)	(3)
Panel A: Bank-firm level			
Post×Treated	0.0462***	–0.0004	0.0003
	(0.0076)	(0.0020)	(0.0019)
Industry×Size×Month FE	Yes	Yes	Yes
Firm×Bank FE	Yes	Yes	Yes
Bank×Month FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Obs.	552,800	1,872,882	1,899,631
Pseudo R-Squared	0.76	0.95	0.95
Panel B: Firm level			
Post×Treated	0.0247***	–0.0018	–0.0015
	(0.0090)	(0.0019)	(0.0019)
Industry×Size×Month FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Obs.	286,954	971,305	971,305
Pseudo R-Squared	0.88	0.98	0.98

same specification after aggregating credit outcomes at the firm-month level.[30] Table 3, Panel B, reports these results.

The firm-level evidence closely mirrors the bank-firm results. Column (1) shows that invoice credit increases by 2.50%, again statistically significant at the 1% level.[31] By contrast, columns (3) and (2) show no significant effect on total credit or non-invoice-based credit, with implied effects of -0.15% and -0.18%, respectively. Taken together, these results support the first mechanism discussed above: eInvoicing increases the use of invoice-based lending. At the same time, they provide little evidence that this average increase in invoice credit translates into broader credit growth or substitution with other types of credit.

4.3 Heterogeneity Analysis: Invoice vs Non-Invoice Firms

The estimates reported in Table 3, however, may conceal substantial heterogeneity across firms. First, the reform may matter more for firms already relying on invoice credit because it improves an existing financing technology. Second, if invoice data improve lenders' broader assessment of borrower quality, credit effects should be stronger precisely for firms for which invoices are more relevant to screening and monitoring. We explore this heterogeneity by re-estimating Equation (1) on two firm subsamples, based on their pre-reform use of invoice credit as a proxy for the relevance of invoices to their financing. One limitation of our approach is that we cannot estimate PPML specifications with invoice credit as the dependent variable for the subsample of non-invoice firms, as they always display zero invoice credit in the pre-reform period.[32]

Table 4 reports these results. The estimates reveal marked differences between invoice and non-invoice firms. Panel A of Table 4 shows that, for invoice firms in treated provinces,

30. In this firm-level specification, we include the same set of time-varying firm controls as Equation (1), as well as the share of firms' credit guaranteed by ICO, an indicator for default status with any bank, and the average of bank-level controls weighted by each bank's credit with the firm. These controls include bank profitability, liquidity, size, non-performing loan ratio, and capitalization. We also include industry×size-tercile×month and firm fixed effects.

31. Because the bank-firm and firm-level specifications differ in their unit of observation and aggregation, the respective coefficient magnitudes are not directly comparable.

32. This follows directly from our definition of non-invoice firms. With no positive pre-treatment observations, the high-dimensional PPML estimator cannot use within-firm or within-relationship variation to identify a treatment effect on invoice credit.

Table 4. eInvoicing and Credit Supply: Invoice vs Non-Invoice Firms

This table reports estimates of β from regressions estimated separately for invoice firms and non-invoice firms over our main sample from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. Invoice firms are firms with positive invoice-based credit in the pre-reform period; non-invoice firms are all other firms. In Panel A, the unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

In Panel B, the unit of observation is a firm-month, credit outcomes are aggregated across all banks lending to firm f in month t, and we estimate:

$$\mathbb{E}[Y_{f,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_f + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_{\bar{b}} \cdot \bar{X}_{b,f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

In Panel A, $Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. In Panel B, $Y_{f,t}$ is the corresponding firm-level amount. For invoice firms, the dependent variable is invoice-based credit in column 1, non-invoice credit in column 2, and total credit in column 3. For non-invoice firms, the dependent variable is total credit in column 4. We cannot estimate specifications using invoice-based credit as the dependent variable for non-invoice firms because, by construction, these firms have no positive invoice-based credit in the pre-reform period. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. Panel A includes industry×size-tercile×month, firm×bank, and bank×month fixed effects. Panel B includes industry×size-tercile×month and firm fixed effects. In both panels, $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, and age. In Panel A, this vector also includes bank-firm-level controls: the share of bank-firm credit guaranteed by ICO and an indicator for whether the firm is in default with that bank. In Panel B, these two variables are replaced by their firm-level counterparts: the share of the firm's total credit guaranteed by ICO and an indicator for whether the firm is in default with any bank. Average bank controls in Panel B are denoted by $\bar{X}_{b,f,t-1}$ and are computed as firm-level weighted averages, with weights based on each bank's share of the firm's credit. These controls include bank profitability, liquidity, size, non-performing loan ratio, and capitalization. In both panels, $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. The row *Controls* indicates that all specifications include firm and Covid controls in Panel A, and firm, Covid, and average bank controls in Panel B. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice Firms			**Non-Invoice Firms**
	Invoice	Non-Invoice	Total	Total
	(1)	(2)	(3)	(4)
Panel A: Bank-firm level				
Post×Treated	0.0407***	0.0028	0.0064**	–0.0070***
	(0.0074)	(0.0032)	(0.0030)	(0.0022)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm×Bank FE	Yes	Yes	Yes	Yes
Bank×Month FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	499,095	714,300	739,956	1,159,022
Pseudo R-Squared	0.75	0.93	0.93	0.97
Panel B: Firm level				
Post×Treated	0.0215**	0.0094***	0.0112***	–0.0143***
	(0.0089)	(0.0026)	(0.0023)	(0.0024)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	229,352	231,015	231,015	739,157
Pseudo R-Squared	0.88	0.98	0.98	0.98

eInvoicing increases total credit by 0.64% and invoice credit by 4.15% relative to invoice firms in control provinces, with a positive, although statistically insignificant, effect on non-invoice credit. For non-invoice firms in treated provinces, by contrast, total credit falls by 0.70% relative to non-invoice firms in the control group.

We then aggregate the data at the firm level to assess whether these relationship-level effects translate into changes to overall firm borrowing. Panel B shows that they do. For invoice firms, total credit rises by 1.13%, invoice credit by 2.17%, and non-invoice credit by 0.94%, with all three effects statistically significant. For non-invoice firms, total credit falls by 1.42%. These estimates imply that the zero average effect on total credit in the full sample masks substantial heterogeneity across firms.

Taken together, the results in Table 4 help interpret the full-sample evidence in Table 3. Firms already using this type of financing before the reform drive the positive average effect on invoice credit. In contrast, the zero average effect on total credit reflects opposite firm-level responses across invoice and non-invoice firms. The positive total-credit effect for invoice firms is also consistent with the idea that improved invoice records support lenders' broader credit risk assessment, but only when invoices are especially informative for screening and monitoring. The negative effect for non-invoice firms suggests a capacity-constraint channel in information use: better information availability for some firms leads to resources being reallocated to those firms, at the cost of fewer resources for firms that do not experience such an increase.

These estimates, however, do not tell us whether the increase in invoice credit reflects larger amounts among existing credit relationships or the creation of new invoice-based credit relationships. We study these margins next.

4.4 Intensive vs Extensive Margin

We now investigate whether eInvoicing changes the amount of credit granted to firms through: i) an increase in credit of pre-existing lending relationships of a given type, akin to an intensive-margin effect; or ii) through an increase in the probability of originating new relationships of a given loan type (invoice-based or non-invoice-based), akin to an

extensive-margin effect. Because both questions concern firms' overall use of different financing instruments across lenders, we conduct this analysis at the firm level.

We begin with the intensive margin. We re-estimate our baseline firm-level specification as in Table 3, Panel B, on a restricted subsample, using total, invoice, and non-invoice credit as dependent variables. Each variable is constructed by aggregating at the firm level all firm-bank relationships with positive pre-reform credit of the corresponding type. To illustrate, consider the following example: If a firm used invoice credit with banks A and C before the reform, but not with bank B, the intensive-margin specification tracks post-reform invoice credit granted through the pre-existing invoice-based lending relationships with banks A and C, aggregated at the firm level. Therefore, we disregard invoice credit that might be granted by bank B after the reform, because the credit relationship with bank B did not previously involve invoice-based financing. In this sense, the intensive-margin estimates capture whether the reform changes credit amounts within pre-existing relationships of a given financing type.

Panel A of Table 5 shows that, in the full sample, the intensive-margin response is concentrated in invoice credit. After splitting the sample into invoice and non-invoice firms, the pattern becomes clearer. For invoice firms, invoice credit rises by 5.79% in relationships already involving invoice credit before the reform. Non-invoice credit increases by 1.25% in relationships involving non-invoice credit before the reform. Total credit also increases by 1.84% among existing relationships. By contrast, for non-invoice firms, total credit falls by 1.22%, and non-invoice credit falls by 1.18%.[33] These estimates indicate that the credit response due to eInvoicing operates in part through changes in credit amounts within pre-existing relationships.

We then turn to the extensive margin. The question is whether eInvoicing affects the probability that a firm originates a new relationship of a given credit type. Because the formation of new relationships is generally infrequent and lumpy, we aggregate the data

33. By construction, we cannot estimate the intensive-margin response for invoice credit for non-invoice firms.

at the firm-quarter level and estimate:

$$Y_{f,q} = \alpha_{i,s,q} + \alpha_{p,i,s} + \beta \cdot Post_q \times Treat_p + \Gamma_f \cdot X_{f,q-1} + \Gamma_{\bar{b}} \cdot \bar{X}_{b,f,q-1} + \Gamma_p \cdot X_{p,i,q} + \varepsilon_{f,q}, \quad (2)$$

where $Y_{f,q}$ is an indicator equal to one if firm f originates a new total, invoice, or non-invoice loan in quarter q with any bank with which the firm did not have that type of loan relationship before. Returning to the invoice-credit example, the extensive margin captures whether, after the reform, a firm is more likely to originate a new invoice loan with a bank that previously did not extend invoice credit to that firm. The industry×size-tercile×quarter fixed effects control for time-varying credit demand factors common to firms of similar size in the same industry, while the province×industry×size fixed effects absorb persistent local differences in the propensity to originate new loans. We also include the same controls as in our baseline firm-level specification. Panel B of Table 5 reports the results. We find that invoice firms are more likely to establish a new invoice relationship, at the 10% significance level. By contrast, non-invoice firms show no meaningful extensive-margin effect.

Taken together, these results suggest that the increase in invoice credit for invoice firms reflects both an intensive-margin effect, through larger credit amounts within pre-existing invoice relationships, and an extensive-margin effect, through a greater likelihood of originating new invoice-based relationships. For non-invoice firms, the decline in credit appears to operate mainly through the intensive margin rather than through reduced origination of new relationships.

4.5 Additional Results on Credit

This section presents additional evidence to interpret the credit results and assess several alternative explanations. We first distinguish between the voluntary and mandatory phases of TicketBAI to study how the credit effects evolve over the course of the reform implementation. We then show that the credit effects documented above are not confined to bank relationships that already used invoice-based financing before the reform, but instead appear to operate more broadly at the firm level. We then show that the

Table 5. eInvoicing and Credit Supply: Intensive and Extensive Margins

This table reports estimates of the effect of eInvoicing on intensive and extensive credit margins over the main sample from January 2021 to December 2024. Panel A reports intensive-margin estimates, where the dependent variables are credit amounts conditional on the firm having positive pre-reform credit of the corresponding type, denoted by $y^{pre} > 0$. Panel A is estimated using Poisson Pseudo-Maximum Likelihood at the firm-month level:

$$\mathbb{E}[Y_{f,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_f + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_{\bar{b}} \cdot \bar{X}_{b,f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

Panel B reports extensive-margin estimates, where the dependent variables are indicators for new credit of the corresponding type, multiplied by 100, aggregated at the firm-quarter level. Panel B is estimated using OLS:

$$Y_{f,q} = \alpha_{i,s,q} + \alpha_{p,i,s} + \beta \cdot Post_q \times Treat_p + \Gamma_f \cdot X_{f,q-1} + \Gamma_{\bar{b}} \cdot \bar{X}_{b,f,q-1} + \Gamma_p \cdot X_{p,i,q} + \varepsilon_{f,q}.$$

Columns 1–3 report estimates for the full sample, columns 4–6 for invoice firms, and columns 7–9 for non-invoice firms. Within each group, the dependent variables are total credit, invoice-based credit, and non-invoice credit, respectively. Column 8 in Panel A is not estimated because non-invoice firms do not have positive pre-reform invoice credit by construction. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. In Panel A, $X_{f,t-1}$ denotes previous-year firm controls, which include firm size, leverage, liquidity, tangibility, profitability, age, the share of the firm's total credit guaranteed by ICO, and an indicator for whether the firm is in default with any bank. Average bank controls are denoted by $\bar{X}_{b,f,t-1}$ and are computed as firm-level weighted averages, with weights based on each bank's share of the firm's credit. These controls include bank profitability, liquidity, size, non-performing loan ratio, and capitalization. Covid controls are denoted by $X_{p,i,t}$ and include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. Panel B includes the corresponding quarterly controls. The row *Controls* indicates that all specifications in every panel include firm, Covid, and average bank controls. All variables are defined in Table IA.I. Standard errors are reported in parentheses and are clustered at the province×month level in Panel A and at the province×quarter level in Panel B. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Full Sample			**Invoice Firms**			**Non-Invoice Firms**		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Intensive margin, conditional on $y^{pre} > 0$									
Dependent variable	Total	Invoice	Non-Invoice	Total	Invoice	Non-Invoice	Total	Invoice	Non-Invoice
Post×Treated	0.0026	0.0564***	–0.0007	0.0182***	0.0563***	0.0124***	–0.0123***	–	–0.0119***
	(0.0027)	(0.0129)	(0.0033)	(0.0034)	(0.0129)	(0.0047)	(0.0035)	–	(0.0035)
Industry×Size×Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–	Yes
Obs.	966,532	156,328	964,560	230,807	153,326	228,808	734,592	–	734,580
Pseudo R-Squared	0.99	0.93	0.98	0.98	0.93	0.98	0.98	–	0.98
Panel B: Extensive margin									
Dependent variable	New Total	New Invoice	New Non-Invoice	New Total	New Invoice	New Non-Invoice	New Total	New Invoice	New Non-Invoice
Post×Treated	0.0200	0.0779	0.0079	0.2562	0.7091*	0.0580	–0.0123	–0.0438	0.0017
	(0.1206)	(0.1002)	(0.0963)	(0.3365)	(0.3971)	(0.2489)	(0.1141)	(0.0369)	(0.1139)
Industry×Size×Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province×Industry×Size FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	321,964	321,964	321,964	76,865	76,865	76,865	244,717	244,717	244,717
R-Squared	0.02	0.03	0.02	0.04	0.04	0.04	0.03	0.03	0.03

main findings are unchanged when we employ a within-province analysis. Finally, we address the concern that our results may reflect newly reported sales due to improved tax compliance rather than better informational properties of invoices. Additional robustness checks are reported in Internet Appendix C.

4.5.1 Voluntary vs Mandatory Periods.

In our baseline specification, we conservatively treat the start of the voluntary adoption phase as the beginning of the treatment period, as we do not directly observe firm-level compliance. To better understand the timing of the reform's effects, we now estimate a

specification that distinguishes between the voluntary and mandatory adoption phases. Because this exercise exploits province-specific treatment timing in a staggered fashion, it also allows us to include Gipuzkoa, which we exclude from the baseline analysis because its voluntary phase begins with the start of our sample.

Table 6 reports the results. For invoice firms, the effects already begin to emerge during the voluntary phase: the coefficients on invoice credit, non-invoice credit, and total credit are generally positive relative to the control group. The mandatory phase then adds a further positive and statistically significant effect across all three credit outcomes, at both the bank-firm and firm levels. For non-invoice firms, by contrast, the voluntary phase has no meaningful effect, whereas the mandatory phase is associated with a significant decline in total credit.

Overall, these results suggest that the credit effects of eInvoicing begin to materialize during the voluntary phase and strengthen once adoption becomes mandatory, with negative effects for non-invoice firms appearing only in the latter period. This corroborates our baseline treatment definition and alleviates the concern that province-specific developments other than eInvoicing drive our results.

4.5.2 Within-Firm Analysis

A natural question is whether the increase in invoice credit documented above reflects a shift toward specific lending relationships within the same firm, rather than an effect operating at the firm level. To address this, we estimate the same specification as in Equation (1), including firm×month fixed effects. This within-firm specification compares, for the same firm in the same month, bank relationships already associated with invoice use in the pre-reform period ("invoice relationships") with those that were not. Since we require at least one credit relationship with invoice-based lending, this analysis focuses only on invoice firms.

Table 7 shows no differential post-reform increase in either total credit or non-invoice credit for pre-existing invoice relationships relative to other relationships of the same firm. This suggests that the effects we document in our main analysis are not driven by

Table 6. eInvoicing and Credit Supply: Voluntary vs. Mandatory Periods

This table reports estimates from staggered difference-in-differences regressions over the main sample, including firms headquartered in Gipuzkoa, from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. In Panel A, the unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta_V \cdot \textit{Treated(Voluntary)}_{p,t} + \beta_M \cdot \textit{Treated(Mandatory)}_{p,t} + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

In Panel B, the unit of observation is a firm-month, credit outcomes are aggregated across all banks lending to firm f in month t, and we estimate:

$$\mathbb{E}[Y_{f,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_f + \beta_V \cdot \textit{Treated(Voluntary)}_{p,t} + \beta_M \cdot \textit{Treated(Mandatory)}_{p,t} + \Gamma_f \cdot X_{f,t-1} + \Gamma_{\bar{b}} \cdot \bar{X}_{b,f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

In Panel A, $Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. In Panel B, $Y_{f,t}$ is the corresponding firm-level amount. The dependent variable is invoice-based credit in column 1, non-invoice credit in column 2, and total credit in column 3 for invoice firms; for non-invoice firms, the dependent variable is total credit in column 4. We cannot estimate specifications with invoice-based credit as the dependent variable for non-invoice firms because, by construction, these firms have no positive invoice-based credit in the pre-reform period. $\textit{Treated(Voluntary)}_{p,t}$ is an indicator equal to one for treated provinces once firms in province p enter the voluntary eInvoicing regime. $\textit{Treated(Mandatory)}_{p,t}$ is an additional indicator equal to one for treated provinces once eInvoicing becomes mandatory in province p. Therefore, β_V measures the effect of entering the voluntary regime relative to the pre-reform period and relative to the control group, while β_M measures the incremental effect of the mandatory regime relative to the voluntary regime and relative to the control group. Panel A includes industry×size-tercile×month, firm×bank, and bank×month fixed effects. Panel B includes industry×size-tercile×month and firm fixed effects. In both panels, $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, and age. In Panel A, this vector also includes bank-firm-level controls: the share of bank-firm credit guaranteed by ICO and an indicator for whether the firm is in default with that bank. In Panel B, these two variables are replaced by their firm-level counterparts: the share of the firm's total credit guaranteed by ICO and an indicator for whether the firm is in default with any bank. Average bank controls in Panel B are denoted by $\bar{X}_{b,f,t-1}$ and are computed as firm-level weighted averages, with weights based on each bank's share of the firm's credit. These controls include bank profitability, liquidity, size, non-performing loan ratio, and capitalization. In both panels, $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. The row *Controls* indicates that all specifications include firm and Covid controls in Panel A, and firm, Covid, and average bank controls in Panel B. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice Firms			**Non-Invoice Firms**
	Invoice	Non-Invoice	Total	Total
	(1)	(2)	(3)	(4)
Panel A: Bank-firm level				
Treated(Voluntary)	0.0283***	0.0088***	0.0095***	0.0013
	(0.0073)	(0.0028)	(0.0024)	(0.0021)
Treated(Mandatory)	0.0399***	0.0052*	0.0067***	–0.0122***
	(0.0060)	(0.0027)	(0.0025)	(0.0021)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm×Bank FE	Yes	Yes	Yes	Yes
Bank×Month FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	625,587	889,893	922,054	1,421,398
Pseudo R-Squared	0.75	0.93	0.93	0.97
Panel B: Firm level				
Treated(Voluntary)	0.0042	0.0106***	0.0085***	–0.0014
	(0.0080)	(0.0022)	(0.0022)	(0.0026)
Treated(Mandatory)	0.0323***	0.0038**	0.0064***	–0.0162***
	(0.0063)	(0.0019)	(0.0020)	(0.0028)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	288,128	289,683	289,683	911,054
Pseudo R-Squared	0.88	0.98	0.98	0.98

relationships that were already used for invoice-based financing before the reform. Rather, they appear to reflect a broader firm-level effect, not one confined to a particular subset of lending relationships.

Table 7. eInvoicing and Credit Supply: Within-Firm Analysis

This table reports estimates of β from regressions over the sample of invoice firms from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. The unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{f,t} + \alpha_{f,b} + \alpha_{b,t} + \gamma \cdot Post_t \times Invoice\ Rel_{f,b} + \beta \cdot Post_t \times Treat_p \times Invoice\ Rel_{f,b} + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

$Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. The dependent variable is total credit in column 1 and non-invoice credit in column 2. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. $Invoice\ Rel_{f,b}$ is an indicator equal to one for bank-firm relationships with positive invoice-based credit in the pre-reform period. The specification includes firm×month, firm×bank, and bank×month fixed effects. $X_{f,t-1}$ and $X_{p,i,t}$ denote the same vectors of previous-year firm controls (including the bank-firm-level ICO share and default indicator) and Covid controls as in Table 3. The row *Controls* indicates that all specifications include firm and Covid controls; the row *Other Interaction Terms* indicates that the lower-order interaction $Post_t \times Invoice\ Rel_{f,b}$ is included (the remaining lower-order terms are absorbed by the fixed effects). All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice Firms	
	Total	Non-Invoice
	(1)	(2)
Post×Treated×Invoice Relationship	0.0012	–0.0063
	(0.0067)	(0.0061)
Firm×Month FE	Yes	Yes
Firm×Bank FE	Yes	Yes
Bank×Month FE	Yes	Yes
Controls	Yes	Yes
Other Interaction Terms	Yes	Yes
Obs.	651,678	627,575
Pseudo R-Squared	0.95	0.95

4.5.3 Within-Province Analysis

One concern is that the main estimates may be driven by province-level shocks that differentially affect treated and control provinces over time. To address this, we estimate a triple-difference specification that augments the baseline with an interaction distinguishing invoice from non-invoice firms. This design allows us to include province×industry×size-tercile×month fixed effects, which absorb shocks common to firms within the same province, industry, size class, and month. This controls not only for aggregate macroeconomic and industry-size-time variation, but also for time-varying province-specific factors such as

regional economic conditions, COVID-19-related policies, local labor-market dynamics, and other fiscal or regulatory changes that may affect all firms in a given local cell. Because the $Post_t \times Treat_p$ interaction varies only at the province-month level, it is absorbed by this fixed-effects structure. Identification thus comes only from whether invoice firms respond differently from non-invoice firms within the same province×industry×size-tercile×month cell. The coefficient on $Post_t \times Treat_p \times Invoice\ firm_f$ is thus identified under the weaker assumption that, absent the reform, invoice and non-invoice firms in treated provinces would have evolved similarly relative to their counterparts in control provinces, conditional on this saturated set of local-sectoral-time controls. Table 8 shows that this coefficient remains positive and statistically significant at both the bank-firm and firm level, indicating that the relative expansion in credit for invoice firms is not explained by province-level time-varying shocks shared by all firms in the same local economic environment.

4.5.4 Tax Compliance and Reporting Incentives.

One concern is that our estimates may reflect improved tax compliance rather than better informational properties of invoices. Since eInvoicing is designed precisely to reduce underreporting and strengthen tax enforcement, previously hidden sales may emerge after the reform. This would mechanically increase the stock of pledgeable receivables and, in turn, invoice-based borrowing. Evidence suggests that eInvoicing adoption can improve tax compliance (e.g., Bellon et al. 2022). At the same time, this mechanism is not an obvious explanation for our credit results, because banks care about borrowers' true repayment capacity, not only about what is formally reported to the tax authority (Artavanis, Morse, and Tsoutsoura 2016).

In our setting, however, no reliable granular firm- or industry-level measures of tax evasion would allow us to test this channel directly. We therefore examine whether eInvoicing adoption induced higher reported firm sales in treated provinces relative to the control group. The evidence in Figure 1, where we re-estimate our main specification at the firm level using log revenues as the dependent variable, shows that this is not the case. As a more direct test, we scale invoice-based credit by firm revenues and re-estimate our baseline spec-

Table 8. eInvoicing and Credit Supply: Within-Province Analysis

This table reports estimates from triple-difference regressions over the main sample from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. The regressions compare the effect of eInvoicing on invoice firms relative to non-invoice firms. In Panel A, the unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{\mathcal{F}} + \gamma \cdot Post_t \times Invoice\ Firm_f + \beta_1 \cdot Post_t \times Treat_p + \beta_2 \cdot Post_t \times Treat_p \times Invoice\ Firm_f + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

In Panel B, the unit of observation is a firm-month, credit outcomes are aggregated across all banks lending to firm f in month t, and we estimate:

$$\begin{aligned}\mathbb{E}[Y_{f,t} \mid \cdot] = \exp\{&\alpha_{\mathcal{F}} + \gamma \cdot Post_t \times Invoice\ Firm_f + \beta_1 \cdot Post_t \times Treat_p \\ &+ \beta_2 \cdot Post_t \times Treat_p \times Invoice\ Firm_f + \Gamma_f \cdot X_{f,t-1} + \Gamma_{\bar{b}} \cdot \bar{X}_{b,f,t-1} + \Gamma_p \cdot X_{p,i,t}\}.\end{aligned}$$

In Panel A, $Y_{f,b,t}$ is the total credit extended by bank b to firm f in month t. In Panel B, $Y_{f,t}$ is firm-level total credit. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. $Invoice\ Firm_f$ is an indicator equal to one for firms with positive invoice-based credit in the pre-reform period. The coefficient β_2 estimates whether the effect of eInvoicing differs between invoice and non-invoice firms, comparing treated and control provinces before and after the reform. The term $\alpha_{\mathcal{F}}$ denotes the fixed effects included in each specification. Column 1 includes industry×size-tercile×month fixed effects, while column 2 includes province×industry×size-tercile×month fixed effects. Panel A additionally includes firm×bank and bank×month fixed effects; Panel B additionally includes firm fixed effects. In both panels, $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, and age. In Panel A, this vector also includes bank-firm-level controls: the share of bank-firm credit guaranteed by ICO and an indicator for whether the firm is in default with that bank. In Panel B, these two variables are replaced by their firm-level counterparts: the share of the firm's total credit guaranteed by ICO and an indicator for whether the firm is in default with any bank. Average bank controls in Panel B are denoted by $\bar{X}_{b,f,t-1}$ and are computed as firm-level weighted averages, with weights based on each bank's share of the firm's credit. These controls include bank profitability, liquidity, size, non-performing loan ratio, and capitalization. In both panels, $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. In column 2, Covid controls are absorbed by province×industry×size-tercile×month fixed effects. The row *Controls* indicates that all specifications include firm and Covid controls in Panel A, and firm, Covid, and average bank controls in Panel B. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Total	
	(1)	(2)
Panel A: Bank-firm level		
Post×Treated	–0.0129***	
	(0.0027)	
Post×Treated×Invoice Firm	0.0261***	0.0273***
	(0.0045)	(0.0062)
Industry×Size×Month FE	Yes	No
Province×Industry×Size×Month FE	No	Yes
Firm×Bank + Bank×Month FEs	Yes	Yes
Controls	Yes	Yes
Other Interaction Terms	Yes	Yes
Obs.	1,899,631	1,896,392
Pseudo R-Squared	0.95	0.96
Panel B: Firm level		
Post×Treated	–0.0112***	
	(0.0022)	
Post×Treated×Invoice Firm	0.0199***	0.0100**
	(0.0036)	(0.0045)
Industry×Size×Month FE	Yes	No
Province×Industry×Size×Month FE	No	Yes
Firm FE	Yes	Yes
Controls	Yes	Yes
Other Interaction Terms	Yes	Yes
Obs.	971,305	964,770
Pseudo R-Squared	0.98	0.99

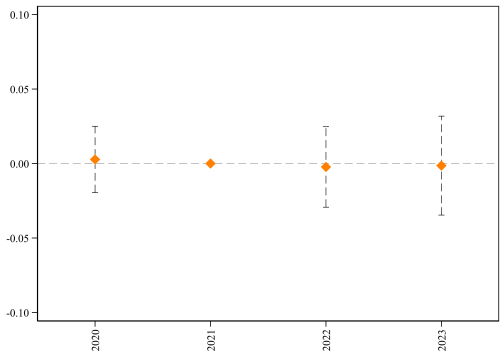
ifications (see Table IA.V in Internet Appendix C.1). If the increase in invoice credit merely reflected newly reported sales, invoice credit would not rise relative to revenues. Instead, invoice-based credit increases relative to revenues. In addition, in our triple-difference specification, we account for province×industry×size-tercile×time fixed effects (see Table 8). To the extent that firms of similar size, in the same province, in the same industry, and in the same month have similar scope for underreporting, these fixed effects attenuate concerns that differential changes in tax evasion drive our results. Overall, we conclude that our results are unlikely to be driven only by newly reported sales or increased tax compliance.

Figure 1. DiD Coefficients of Log Revenues

This figure plots estimates of β_ℓ from a dynamic difference-in-differences specification estimated for the average firm in our main sample from 2020 to 2023 using OLS.

$$Y_{f,t} = \alpha_{i,s,t} + \alpha_f + \sum_{\ell \neq -1} \beta_\ell \cdot \mathbf{1}\{t - T_p = \ell\} \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \varepsilon_{f,t}.$$

$Y_{f,t}$ is the logarithm of revenues of firm f in year t. $\mathbf{1}\{t - T_p = \ell\}$ is an indicator equal to one when year t is ℓ periods away from the treatment year T_p of province p, with $\ell = -1$ omitted as the reference period. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. The specification includes industry×size-tercile×year fixed effects ($\alpha_{i,s,t}$) and firm fixed effects (α_f). $X_{f,t-1}$ denotes the vector of lagged firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, and age. All variables are defined in Table IA.I. Confidence intervals are at the 95% level.



4.6 Interest Rates

The evidence presented so far demonstrates that the introduction of eInvoicing has significant and heterogeneous effects on credit quantities, but these results do not, by themselves, allow us to distinguish between supply-side and demand-side explanations, as they reflect equilibrium outcomes. The observed changes in invoice credit, non-invoice credit, and total credit following the introduction of TicketBAI could, in principle, reflect

shifts in firms' demand for different forms of borrowing due to the implementation, or also changes in lenders' willingness to provide them. Analyzing the joint effects of the introduction on prices (loan rates) and quantities helps to distinguish between these two interpretations. For invoice and non-invoice credit, if changes in credit supply drive the observed increase (decrease) in quantities, treated firms should obtain more (less) credit at lower (higher) interest rates. By contrast, if changes in credit demand are the primary drivers, they should move both prices and quantities in the same direction.

We begin by examining the effects of eInvoicing on the cost of invoice and non-invoice credit. This helps assess whether the quantity effects are consistent with a pure demand explanation or whether the evidence suggests that changes in credit supply also play a significant role. We then examine the overall cost of credit. Formally, we estimate the same specification as in Equation (1) using OLS at the bank-firm-month level from January 2021 to December 2024, replacing the dependent variable with the logarithm of the cost of credit, measured as the credit-weighted average interest rate on outstanding credit:

$$\log(R_{f,b,t}) = \alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_c \cdot C_{f,b,t} + \Gamma_p \cdot X_{p,i,t} + \varepsilon_{f,b,t} \quad (3)$$

where the subscripts follow the same notation as in Equation (1), and the dependent variable $\log(R_{f,b,t})$ denotes the logarithm of the interest rate charged by bank b to firm f in month t, computed separately for total credit, invoice credit, and non-invoice credit. As before, β captures the differential change in borrowing costs for firms in the treated provinces after the introduction of eInvoicing relative to firms in control provinces. The fixed effects and control variables are the same as in the baseline credit specification. To account for possible compositional shifts, especially when using the total cost of credit, we also include the share of fixed-rate credit, the share of term loans, the share of credit guaranteed by physical collateral, the weighted-average maturity, and bank credit relative to firms' total assets. Standard errors are clustered at the province×month level.

Table 9 reports the effects of eInvoicing on borrowing costs. We begin with invoice and non-invoice credit, since these prices are the most informative for interpreting the quantity results. For invoice firms in treated provinces, the cost of invoice credit declines

Table 9. eInvoicing and Interest Rates

This table reports estimates of β from interest-rate regressions estimated separately over the subsamples of invoice firms and non-invoice firms from January 2021 to December 2024, using OLS. In Panel A, the unit of observation is a bank-firm-month, and we estimate:

$$\log(R_{f,b,t}) = \alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_c \cdot C_{f,b,t} + \Gamma_p \cdot X_{p,i,t} + \varepsilon_{f,b,t}.$$

In Panel B, the unit of observation is a firm-month, interest rates are aggregated across all banks lending to firm f in month t using credit-volume weights, and we estimate:

$$\log(R_{f,t}) = \alpha_{i,s,t} + \alpha_f + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_{\bar{b}} \cdot \bar{X}_{b,f,t-1} + \Gamma_c \cdot C_{f,t} + \Gamma_p \cdot X_{p,i,t} + \varepsilon_{f,t}.$$

	Invoice Firms			**Non-Invoice Firms**
	Invoice	Non-Invoice	Total	Total
	(1)	(2)	(3)	(4)
Panel A: Bank-firm level				
Post×Treated	–0.0243***	0.0380***	0.0360***	0.0387***
	(0.0056)	(0.0060)	(0.0056)	(0.0059)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm×Bank FE	Yes	Yes	Yes	Yes
Bank×Month FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	86,178	623,614	630,958	888,345
R-Squared	0.88	0.82	0.83	0.82
Panel B: Firm level				
Post×Treated	–0.0213***	0.1005***	0.0936***	0.0913***
	(0.0055)	(0.0092)	(0.0089)	(0.0082)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	68,220	218,634	219,512	581,416
R-Squared	0.85	0.83	0.84	0.80

by approximately 2.4%, while the cost of non-invoice credit rises by approximately 3.9% relative to invoice firms in control provinces. For non-invoice firms, the cost of total credit also rises by 3.9%. The firm-level results show the same pattern.

Taken together with the quantity results, this evidence is consistent with a supply-side component. In equilibrium, the quantity of invoice credit increases while its price decreases. If treated firms simply demanded relatively more invoice credit after the reform, we would instead expect both quantities and prices to rise. At the same time, non-invoice credit becomes more expensive even though its quantity either remains flat or declines relative to the control group, again inconsistent with a purely demand-driven explanation. The

results for the total cost of credit are less direct to interpret because total borrowing costs reflect the composition of borrowing across invoice and non-invoice instruments, which our previous credit quantity results show has changed. In the data, the cost of total credit rises in treated provinces relative to the control group for both invoice and non-invoice firms. This need not contradict the existence of a supply-side component in the quantity results. Invoice credit is normally more expensive on average than other forms of borrowing, so a shift in the composition of credit toward invoice-based instruments can raise the average cost of total credit even if the price of invoice credit itself decreases in relative terms.[34] Overall, the joint behavior of prices and quantities suggests that eInvoicing changes the equilibrium allocation of credit not only through firms' demand, but also through lenders' supply decisions. The next section analyzes more directly whether these results reflect changes in lenders' ability to process information.

5. eInvoicing and Lenders' Use of Information

The evidence in Section 4 shows that eInvoicing reshapes credit allocation, as invoice firms in treated provinces obtain more total and invoice-based credit and face lower costs of invoice-based borrowing than comparable firms in control provinces. One plausible interpretation is that the reform changes the information available to lenders by making invoice records more standardized, verifiable, and usable for lending purposes. If eInvoicing improves the informational content of invoices in ways relevant to credit risk assessment, banks should be able to differentiate more effectively among borrowers and allocate credit accordingly. In this section, we test this hypothesis by analyzing the effects of eInvoicing on two complementary measures of banks' information: the cross-sectional dispersion of interest rates and banks' internal risk assessments, and the accuracy of these internal risk assessments in predicting ex-post defaults.

34. The average interest rate is 4.43% for invoice credit and 3.70% for non-invoice credit, while the average interest rate at the firm level is 3.71%. See Table IA.IV in Internet Appendix B.

5.1 Dispersion

We first focus on the cross-sectional dispersion of both loan rates and banks' internal PDs. When banks rely on a coarse information set, observably similar borrowers tend to receive similar rates and risk assessments, thereby compressing the cross-section. However, when lenders can instead use richer, more informative signals, differences among borrowers become more salient, and prices and PDs can vary more widely across otherwise similar firms (Cornell and Welch 1996). If eInvoicing improves the information available to lenders, banks should be able to tailor rates and internal risk assessments more closely to borrower-specific signals, thereby increasing cross-sectional dispersion (Skrastins and Vig 2019; Cerqueiro, Degryse, and Ongena 2011; Doerr et al. 2023).

5.1.1 Methodology

To test this hypothesis, we study how the dispersion of loan rates and internal risk assessments changes across treated and control provinces over time. Formally, we estimate the following specification at the province-quarter level:

$$\log\left(y_{p,t}\right) = \beta \cdot Post_t \times Treat_p + \alpha_p + \alpha_t + \Gamma \cdot X_{p,t} + \varepsilon_{p,t} \tag{4}$$

where p denotes province and t the year-quarter, and $log(y_{p,t})$ is the log of the interquartile range or the standard deviation of either loan interest rates or banks' internal PDs across all credit relationships in a given province-quarter.[35] The coefficient of interest, β, captures whether cross-sectional dispersion changes differentially in treated provinces relative to the control group after the reform. The vector $X_{p,t}$ includes province-quarter controls meant to absorb differences in local credit-market conditions, borrower composition, and the intensity of COVID-19-related disruptions.[36] Because the informational content of

35. We conduct this analysis at the province-quarter level because PDs are reported only quarterly. Specifically, we compute measures of dispersion for interest rates using only new loans and averaging rates at the bank-firm-quarter level, weighted by loan amounts; for PD dispersion, we consider all outstanding bank-firm relationships, as PDs are updated quarterly. See Internet Appendix D.2 for more details about PDs.
36. Specifically, the province-quarter controls include: the logarithm of total credit, the number of firms, the corporate credit-market HHI, the share of ICO-guaranteed credit, the shares of credit to the main sectors of economic activity – Agriculture, Mining, Manufacturing, Construction, Retail and Wholesale, Transport, and Hospitality – as well as the Google Mobility Index and the total number of workers

eInvoicing is likely most relevant for firms that already relied on invoice-based credit before the reform, we estimate Equation (4) separately for invoice and non-invoice firms.

We then pool the two groups and estimate a specification that allows the treatment effect to differ across firm types while absorbing province×firm-type and firm-type×quarter fixed effects:

$$\begin{aligned}\log(y_{p,g,t}) = \beta_1 \cdot Post_t \times Treat_p + \beta_2 \cdot Post_t \times Treat_p \times Invoice\ Firms_g \\ + \alpha_{p,g} + \alpha_{g,t} + \Gamma_1 \cdot X_{p,t} + \Gamma_2 \cdot X_{p,t} \times Invoice\ Firms_g + \varepsilon_{p,g,t}\end{aligned} \tag{5}$$

where p indexes province, g firm type, and t year-quarter. $X_{p,t}$ is the same vector of controls as in Equation (4). This specification allows us to test directly whether the change in dispersion after the reform differs between invoice and non-invoice firms, after accounting for persistent differences across provinces within each firm type and time-specific shocks common to a given firm type across all provinces. The interaction between $X_{p,t}$ and $Invoice\ Firms_g$ allows the control variables to have different coefficients across invoice and non-invoice groups, making the estimates of Equation (5) comparable to the ones of Equation (4).

5.1.2 Results

Table 10 reports the results of estimating Equation (4) and Equation (5). The estimates reveal marked heterogeneity across firm types, consistent with the credit allocation results documented in Section 4. Across both panels and both dispersion measures, the estimated coefficients point to the same pattern: after the reform, dispersion widens for invoice firms in treated provinces relative to control provinces, while it does not increase for non-invoice firms and, if anything, decreases. This is consistent with lenders differentiating more across invoice firms and less across non-invoice firms after the introduction of eInvoicing. Because the analysis is conducted at the province×quarter level, however, the effective sample size is limited, and the separate estimates for each group of firms are not uniformly statistically significant. The triple-interaction estimates, which directly test the differential

covered by the ERTE program. We construct all province-quarter controls within the estimation sample used in each analysis to compute the dispersion measures.

effect across the two groups, are statistically significant in all four specifications.[37]

Panel A reports the results when using the dispersion of loan interest rates as the dependent variable. Under the interquartile-range measure (columns (1)–(3)), the coefficient in column (1) implies that the dispersion of interest rates increases by $100 \times (\exp 0.16 - 1) = 17\%$ for invoice firms in treated provinces after the reform relative to the control group, an effect that is statistically significant at the 10% level.[38] The coefficient in column (2) implies that dispersion does not increase for non-invoice firms; the point estimate corresponds to a decline of 10%, although it is not statistically significant. The triple interaction in column (3) confirms that the difference between the two groups is economically large and statistically significant at the 5% level. The standard-deviation specifications (columns (4)–(6)) deliver the same qualitative picture, although with more noise: the coefficients in columns (4) and (5) imply an imprecisely estimated increase of 8% for invoice firms and a statistically insignificant decline of 13% for non-invoice firms, while the triple interaction in column (6) confirms that the difference across the two groups remains statistically significant.

Panel B repeats the exercise using the dispersion of banks' internal PDs. Under the interquartile-range measure (columns (1)–(3)), the coefficient in column (1) implies that PD dispersion increases by 14% for invoice firms in treated provinces after the reform relative to invoice firms in control provinces, an effect that is statistically significant at the 5% level. By contrast, the coefficient in column (2) implies that PD dispersion does not increase for non-invoice firms, with a statistically insignificant point estimate corresponding to a decline of 9%. The triple interaction in column (3) again confirms that the difference between the two groups is large and statistically significant at the 1% level. The standard-deviation specifications (columns (4)–(6)) preserve the same sign pattern: the coefficient in column (4) implies a small and statistically insignificant increase of 6% for invoice firms, the coefficient in column (5) implies a decline of 12% for non-invoice firms, significant at the 5% level, and the triple interaction in column (6) confirms that

37. Results are qualitatively similar without taking the log of the dependent variables. See Table IA.XIII in Internet Appendix D.1.

38. Throughout Section 5.1, we report coefficient estimates that have already been transformed into percentage effects, computed as $100 \times \left(\exp \hat{\beta} - 1\right)$.

the difference across the two groups remains statistically significant at the 5% level.

Taken together, these results indicate a marked difference in the effects of eInvoicing across firm types. Across all four specifications, the post-reform change in dispersion in both loan rates and risk estimates is significantly larger for invoice firms than for non-invoice firms. For invoice firms, lenders appear to assign more borrower-specific prices and internal risk assessments after the reform, widening the cross-sectional dispersion of both interest rates and PDs. For non-invoice firms, by contrast, dispersion does not increase, and the point estimates suggest, if anything, a compression, consistent with a more uniform treatment of borrowers whose invoice records are less directly relevant for lending decisions. Overall, these results are consistent with eInvoicing changing the information banks can use in pricing and risk assessment, especially for firms whose financing is more closely tied to invoices.

5.2 Predictive Accuracy of Probabilities of Default

To complement the evidence on dispersion, we next examine the accuracy of banks' internal PDs in predicting ex-post defaults. This is arguably a more direct measure of the quality of banks' information, as it captures banks' ability to produce internal risk assessments that more accurately predict the realization of credit risk.

5.2.1 Methodology

To capture how well banks' PDs can predict borrowers' credit risk, we employ the Area Under the Receiver Operating Characteristic curve (AUROC) as a measure of the quality of banks' information (Howes and Weitzner 2026; Beyhaghi, Howes, and Weitzner 2024). The AUROC quantifies the extent to which banks' PDs discriminate across borrowers with different ex-post default outcomes. Intuitively, it reflects the probability that a defaulting borrower is assigned a higher PD than a non-defaulting borrower. An AUROC of 0.5 indicates no discriminatory power (i.e., predictions are no better than a random draw), whereas a value of 1 reflects perfect rank ordering.

For this analysis, we rely on the PDs that banks employing the IRB approach for

Table 10. Dispersion of Interest Rates and Probabilities of Default

This table reports estimates of the effect of eInvoicing on the dispersion of interest rates and banks' probability-of-default (PD) estimates at the province-quarter level. Panel A uses the log dispersion of interest rates as the dependent variable; Panel B uses the log dispersion of banks' PD estimates. Columns 1–3 measure dispersion using the interquartile range, while columns 4–6 measure dispersion using the standard deviation. Columns 1 and 4 restrict the sample to invoice firms; columns 2 and 5 restrict the sample to non-invoice firms; columns 3 and 6 use the full sample and estimate separate effects for invoice and non-invoice firms. For columns 1, 2, 4, and 5, we estimate:

$$\log(y_{p,t}) = \beta \cdot Post_t \times Treat_p + \alpha_p + \alpha_t + \Gamma \cdot X_{p,t} + \varepsilon_{p,t}.$$

For columns 3 and 6, we estimate the corresponding stacked specification:

$$\log(y_{p,g,t}) = \beta_1 \cdot Post_t \times Treat_p + \beta_2 \cdot Post_t \times Treat_p \times Invoice\ Firms_g + \alpha_{p,g} + \alpha_{g,t} + \Gamma_1 \cdot X_{p,t} + \Gamma_2 \cdot X_{p,t} \times Invoice\ Firms_g + \varepsilon_{p,g,t}.$$

In these equations, y is either the interquartile range or the standard deviation of interest rates or PDs within a province-quarter cell. In the stacked specification, g indexes whether the dispersion measure is computed among invoice firms or non-invoice firms, and $Invoice\ Firms_g$ is an indicator equal to one for the invoice-firms group. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for treated provinces. The vector $X_{p,t}$ includes log total credit, the number of firms, the corporate credit-market HHI, the share of ICO-guaranteed credit, the shares of credit to Agriculture, Mining, Manufacturing, Construction, Retail and Wholesale, Transport, and Hospitality, the Google Mobility index, and the total number of workers covered by the ERTE program, all constructed within the sample used to compute the dispersion measures. The row *Controls* indicates that all specifications include the vector $X_{p,t}$ (interacted with $Invoice\ Firms_g$ in the stacked specification). All variables are defined in Table IA.I. Heteroskedasticity-robust standard errors are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Log(Interquartile Range)			Log(Standard Deviation)		
	Invoice	Non-Invoice	All	Invoice	Non-Invoice	All
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Interest Rates						
Post×Treated	0.16*	–0.10	–0.10	0.08	–0.14	–0.14
	(0.09)	(0.07)	(0.07)	(0.07)	(0.09)	(0.09)
Post×Treated×Invoice Firms			0.26**			0.22*
			(0.11)			(0.11)
Province FE + Quarter FE	Yes	Yes	No	Yes	Yes	No
Province×Invoice FE	No	No	Yes	No	No	Yes
Invoice×Quarter FE	No	No	Yes	No	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	96	96	192	96	96	192
R-Squared	0.80	0.85	0.85	0.73	0.80	0.89
Panel B: Probabilities of Default						
Post×Treated	0.13**	–0.09	–0.09	0.06	–0.13**	–0.13**
	(0.05)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)
Post×Treated×Invoice Firms			0.22***			0.19**
			(0.08)			(0.08)
Province FE + Quarter FE	Yes	Yes	No	Yes	Yes	No
Province×Invoice FE	No	No	Yes	No	No	Yes
Invoice×Quarter FE	No	No	Yes	No	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	96	96	192	96	96	192
R-Squared	0.74	0.73	0.74	0.79	0.66	0.77

capital requirements (IRB banks) must report to the Bank of Spain.[39] Hence, these data are available at the bank-firm-quarter level only for IRB banks and the subset of firms borrowing from at least one of these banks.[40] To construct the AUROC, we compute the Receiver Operating Characteristic (ROC) curve by comparing banks' predicted PDs with one-year-ahead realized defaults. For each possible PD threshold, we classify borrowers as defaulting or non-defaulting depending on whether their PD exceeds the threshold. At each threshold, we compute the true positive rate (the share of actual defaulters correctly classified) and the false positive rate (the share of non-defaulters incorrectly classified as defaulters). Plotting the true positive rate against the false positive rate across all possible thresholds yields the ROC curve. The area under this curve summarizes predictive accuracy as a single statistic, i.e., higher AUROC values indicate better performance of the PDs in distinguishing future defaulters from non-defaulters.

Our objective is to test whether the quality of banks' risk assessments – measured as their ability to rank borrowers by risk – changes differentially in response to the eInvoicing mandate. To this end, we compute the AUROC separately for treated and control firms in the pre-reform and post-reform periods, and compare the resulting values across treated and control provinces before and after the introduction of TicketBAI, in the spirit of a graphical difference-in-differences design. Statistical inference for pairwise AUROC comparisons is based on the nonparametric DeLong test (DeLong, DeLong, and Clarke-Pearson 1988), as in Iyer et al. (2016) and Howes and Weitzner (2026).

5.2.2 Main Results

Figure 2 displays the results of the AUROC-based analysis. Both panels show two curves, one for treated firms and one for control firms. For each group, the relevant metric is the area under the curve (i.e., AUROC), and we compare how this area differs between treated and control firms before and after the reform. The green curve corresponds to firms located in the treated provinces (Biscay and Álava), while the grey curve corresponds

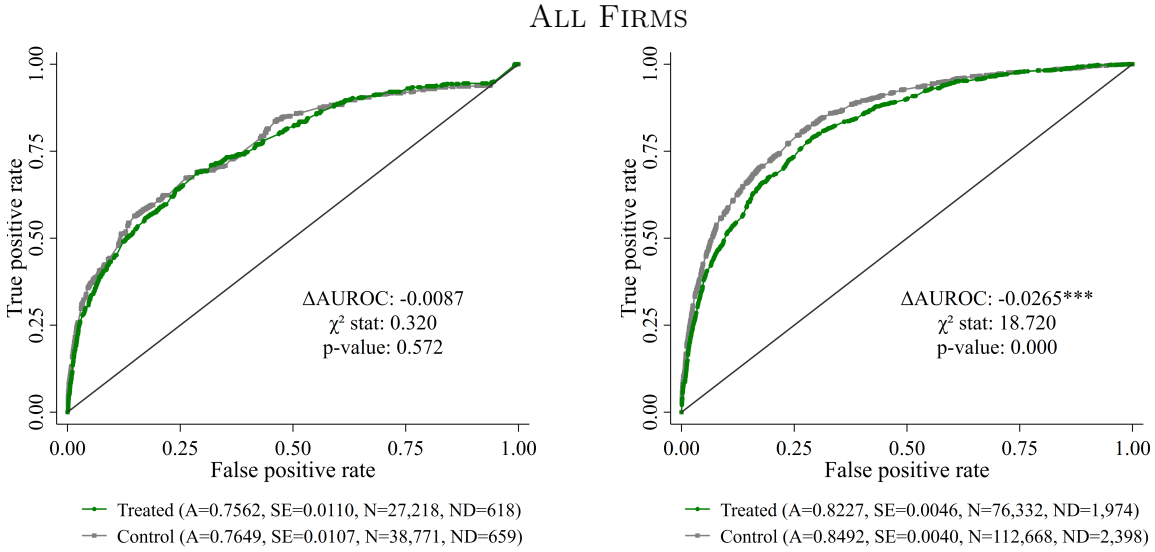
39. In Internet Appendix D.2, we provide additional information about PDs and validate them as a reliable proxy for predicting default.
40. In Internet Appendix C.2 we show that the main results of Section 4 hold for this restricted sample of firms.

to firms located in the control provinces (Burgos, Cantabria, La Rioja, and Navarre). The left panel restricts the sample to bank-firm-quarter observations in 2021, i.e., the period before the introduction of TicketBAI in Biscay and Álava, and the right panel focuses on bank-firm-quarter observations from 2022 to 2024.

Figure 2 shows that the predictive accuracy of banks' internal PDs improved over time in both treated and control provinces. In 2021, before the reform, predictive performance was very similar across the two groups: the difference in AUROC between treatment and control is −0.009 and statistically indistinguishable from zero. In the post-reform period, by contrast, the difference widens to −0.027 and becomes statistically significant at the 1% level. Hence, the implied difference-in-differences is −0.018 and indicates that the improvement in predictive performance was less pronounced for firms in treated provinces. These figures are economically meaningful, as even a 0.01 improvement in AUROCs *"is considered a noteworthy gain in the credit scoring industry"* (Iyer et al. 2016, p. 1565).

Figure 2. Differences in Predictive Quality of PDs before and after eInvoicing

This figure presents the AUROC-based analysis of the predictive quality of the probabilities of default (PD) internally estimated for credit to firms located in treated and control provinces. The larger the area under the curve, the more accurate the estimated PDs. In the legend, A is the AUROC, SE its standard error, N the number of bank-firm-quarter observations, and ND the number of realized defaults. ΔAUROC is the difference between treated and control; the χ^2 statistic and p-value are from the DeLong, DeLong, and Clarke-Pearson (1988) test of equality of the two AUROCs. ***, **, * denote significance at the 1%, 5%, and 10% levels.



(a) 2021 **(b)** 2022–2024

5.2.3 Invoice vs Non-Invoice Firms

As in the previous sections, the aggregate results in Figure 2 mask important heterogeneity across firms. The informational effects of eInvoicing are unlikely to be uniform, since their relevance to lenders should depend on whether invoice records are actually useful for assessing borrower risk. For firms that use invoice-based credit, the reform may improve lenders' internal risk assessments more directly. For firms that do not use invoice-based credit, by contrast, any informational effect is likely more muted and could even be negative if banks reallocate screening and monitoring capacity toward borrowers for whom invoice records are more informative. To examine whether the aggregate pattern in Figure 2 reflects these differences across firms, we split the sample into invoice and non-invoice firms and repeat the AUROC analysis separately for each group, in the spirit of a graphical triple-difference design. Figure 3 reports the results. The top row shows the pre- and post-reform AUROCs for invoice firms in treated and control provinces, while the bottom row reports the corresponding comparison for non-invoice firms.

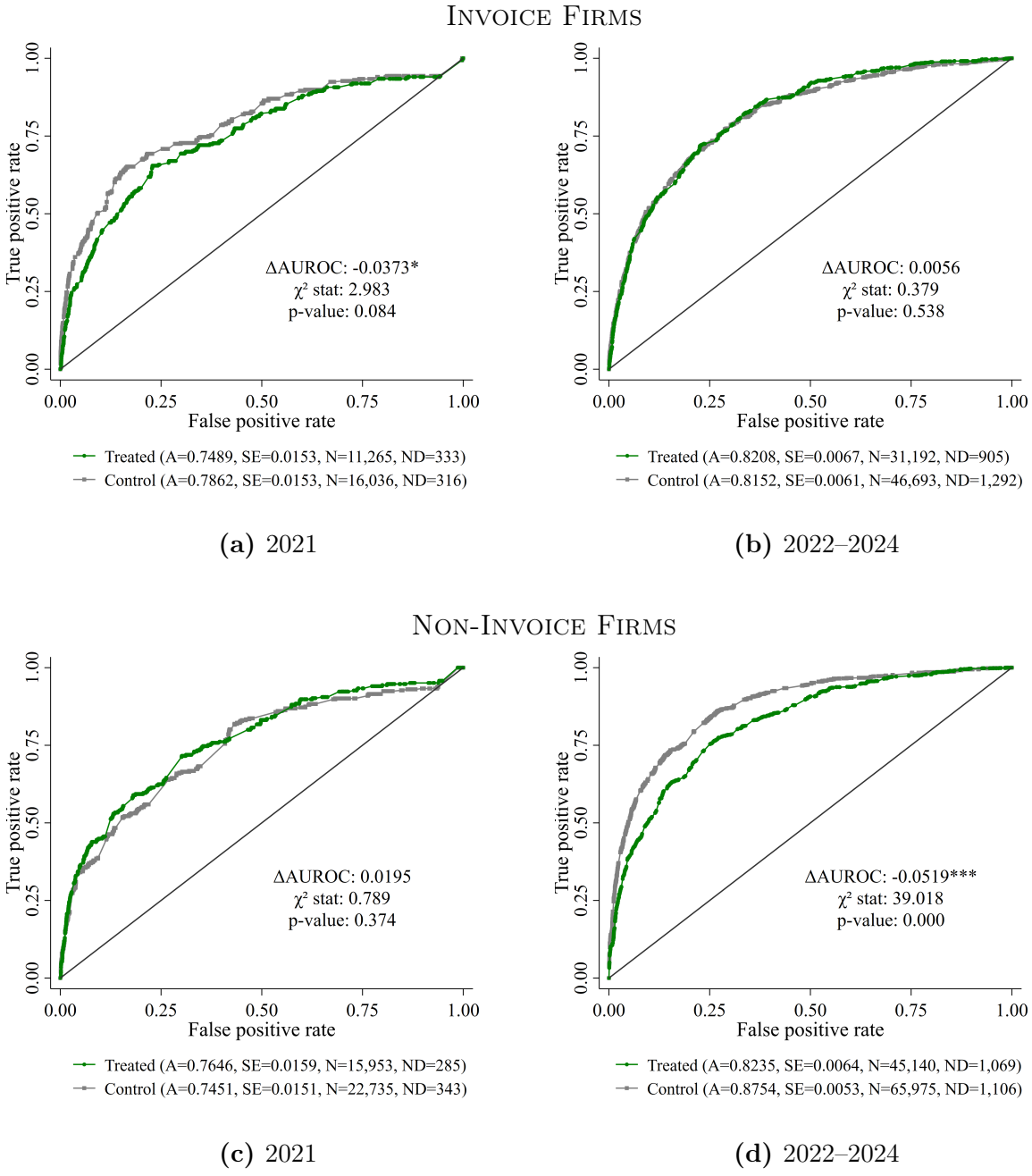
Consistent with our previous results, two patterns emerge depending on firm type. For invoice firms, the treated group exhibits a lower AUROC than the control group in the pre-reform period, with an economically large and statistically significant difference of -0.037. After the reform, this difference shrinks markedly and becomes statistically indistinguishable from zero, with an implied difference-in-differences of 0.043. We interpret this pattern with caution. As treated invoice firms start with lower predictive accuracy than control firms, the post-reform result partly reflects closing a pre-existing gap rather than divergence from a balanced pre-period.[41] With this caveat in mind, the evidence indicates a substantial relative improvement in the predictive accuracy of banks' internal risk assessments for invoice firms in treated provinces.

For non-invoice firms, the opposite pattern emerges. Before the reform, treated and control firms display a difference in AUROCs that is not statistically significant at con-

41. This difference may reflect the fact that, in this graphical exercise, we cannot control for differences in firm characteristics between treated and control provinces. In Section 5.2.4 we present several robustness checks to mitigate this concern.

Figure 3. Differences in Predictive Quality of PDs before and after eInvoicing: Invoice vs Non-Invoice Firms

This figure presents the AUROC-based analysis of the predictive quality of the probabilities of default (PD) internally estimated for credit to firms located in treated and control provinces for invoice and non-invoice firms. The larger the area under the curve, the higher the accuracy of the estimated PDs. Legend, test statistics, and significance levels as in Figure 2.



(a) 2021 **(b)** 2022–2024

(c) 2021 **(d)** 2022–2024

ventional levels. In the post-reform period, the treated curve shifts below the control curve, generating a difference of -0.052 that is statistically significant at the 1% level, making the implied difference-in-differences -0.071. Taken together, these results indicate that the reform is associated with an improvement in banks' ability to predict default for invoice firms, but with a deterioration for non-invoice firms. This result aligns with the evidence on credit allocation in Table 4. Lenders expand credit to invoice firms, for which the information embedded in invoice records appears more relevant for assessing credit risk, while outcomes move in the opposite direction for non-invoice firms.

These findings are consistent with frictions in banks' use of information. The reform appears to increase the informational value of invoice records for invoice firms, but the corresponding decline in predictive accuracy for non-invoice firms suggests that banks may not be able to improve their information across all types of credit. One possible interpretation is that banks adjust their information-processing technologies, rating methodologies, or analysts' attention toward the type of lending for which the reform makes information more valuable. If information-processing capacity is limited, such a reallocation can generate negative spillovers for borrowers or credit products for which invoice data are less informative. This interpretation is consistent with a recent literature emphasizing capacity constraints and limited attention in lending decisions (see, e.g. Campbell, Loumioti, and Wittenberg-Moerman 2019; Mariathasan and Zhuk 2022).

5.2.4 Robustness Tests

The intrinsic design of AUROCs raises two potential concerns. First, we compute AUROC on raw samples and cannot control for additional covariates or fixed effects as we would in a regression framework. Thus, cross-sectional differences in borrower composition and the differential evolution of these characteristics over time across treated and control groups could explain our results. Second, the post-reform window overlaps with the tail end of government-guaranteed (ICO) lending issued during the COVID-19 pandemic and other stabilization policies, including default moratoria. If these programs weakened banks' incentives to produce information, and if their effects differed systematically across treated

and control groups, they could affect the measured change in AUROC for reasons unrelated to eInvoicing. To mitigate these concerns, we conduct a battery of robustness checks.

First, we assess whether cross-sectional differences across treated and control firms drive our results. To this end, in Figure IA.V we re-estimate the AUROCs on a matched sample in which treated and control firms are balanced on observables. We implement the matching procedure in two stages using bank-firm-quarter information from 2021Q4. We first form exact-match cells defined by the firm's industry, size tercile, invoice indicator, and bank identifier, thereby restricting comparisons to highly homogeneous groups of bank–firm pairs. Within each exact-match cell, we then perform nearest-neighbor propensity-score matching based on firm-level financial characteristics, including liquidity, tangibility, and profitability, as well as the bank's IRB PD estimate for the firm. The final matched sample includes only bank-firm-quarter observations for which a match is found in the exact same cell. Table IA.XV shows that this procedure substantially improves the balance of observables across treated and control firms. Figure IA.V shows that matching reduces pre-existing differences in AUROCs, especially for invoice firms, and strengthens the estimated post-reform effects for this group, without materially affecting the results for non-invoice firms.

Second, to mitigate the concern that our results could reflect differential compositional shifts across treated and control provinces, in Figure IA.VI we re-estimate the AUROCs on the subsample of *stayers*, defined as bank–firm relationships observed in both the pre-period and the post-period, so that the comparison is not driven by entry or exit. As an additional check, in Figure IA.VII and Figure IA.VIII, we assess whether the results depend on cohort composition by splitting relationships according to the first observed quarter of each bank–firm relationship. Specifically, we distinguish relationships that begin in 2019 or earlier from those that begin in 2020–2021, ensuring that each cohort contains bank-firm-quarter observations in both the pre- and post-treatment periods. Finally, to rule out that government-guaranteed lending drives our results, in Figure IA.IX, we re-estimate AUROCs after excluding bank-firm-quarter observations in which all outstanding loans are either ICO-guaranteed or under COVID-19 payment moratoria.

Across all of these checks, the main conclusions of Figure 3 remain unchanged: banks'

ability to predict default improves for invoice firms in treated provinces relative to invoice firms in control provinces, and deteriorates for treated non-invoice firms relative to control non-invoice firms.

6. Conclusions

Digitalization is reshaping how firms operate, transact, and obtain external finance. While a large literature has examined the digital execution of payments, finance scholars have largely overlooked digital invoicing. Electronic invoicing changes the properties of invoices by making them standardized, traceable, verifiable, and harder to falsify, thereby altering the information that firms can credibly share with their financiers. With a growing number of jurisdictions, including the European Union, moving toward mandatory eInvoicing as a foundational pillar of their digital strategy, understanding how this reshapes credit markets is a first-order question for both research and policy.

Exploiting the regional rollout of TicketBAI in Spain's Basque Country, together with administrative credit-register and balance-sheet data, this paper provides the first systematic evidence that eInvoicing meaningfully affects firms' financing. Overall, three main findings stand out. First, eInvoicing expands credit for firms already using invoice-based credit, driven by both higher invoice-based credit volumes and a higher likelihood of using such instruments. By contrast, firms that did not rely on invoice-based credit experience a reduction in total credit, with no offsetting increase in the take-up of invoice-related arrangements. Second, the joint behavior of prices and quantities provides suggestive evidence consistent with a supply-side component, with lenders shifting credit provision toward firms already using invoice-based financing, arguably because banks can assess their credit risk better. Indeed, our third main finding suggests that our results are consistent with changes in banks' use of information. Specifically, eInvoicing widens the cross-sectional dispersion of interest rates and internal probabilities of default for firms already using invoice-based credit pre-reform relative to firms that do not. A similar asymmetry appears in the predictive accuracy of banks' risk assessments, which increases for the former and decreases for the latter.

In other words, the evidence is consistent with banks becoming better able to discriminate and predict ex-post default among firms using invoice-based credit. In turn, they redirect credit toward such firms and away from firms whose invoicing information is less relevant.

These results speak to a broader debate on how digitalization is reshaping financial intermediation. A common thread across recent reforms – from open banking, to consumer data rights, to digital identity and unified payment infrastructures – is that policymakers seek to lower the cost of producing and sharing standardized, verifiable digital records about firms and households, with the explicit goal of unlocking better credit decisions and broader financial inclusion.

Our evidence supports the view that such standardization can sharpen lenders' screening and reallocate credit toward borrowers for whom the new digital records are more informative, but it also delivers a cautionary message: the gains from digitalization in credit markets are not uniform. When new digital records are more informative for some borrowers than for others, lenders' improved ability to discriminate generates winners and losers, and the reform may tighten credit precisely for the firms whose activity is least represented in the new data. From a policy perspective, this suggests that the design of eInvoicing and other open-data mandates should be evaluated not only by their average effects but also by their distributional consequences, and that complementary tools – such as broader-coverage data-sharing frameworks, public credit guarantees, or targeted support for under-represented segments – may be useful to mitigate adverse, unintended reallocations during the transition.

We leave to future work several questions that our setting and sample period do not allow us to address fully, including the long-run effects of eInvoicing on firm investment, growth, and exit; whether better invoice information levels the playing field for new entrants; and the real effects that may arise as banks gradually internalize the richer information environment.

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Internet Appendix

for

"Digitalization and Credit Markets: Evidence from eInvoicing"

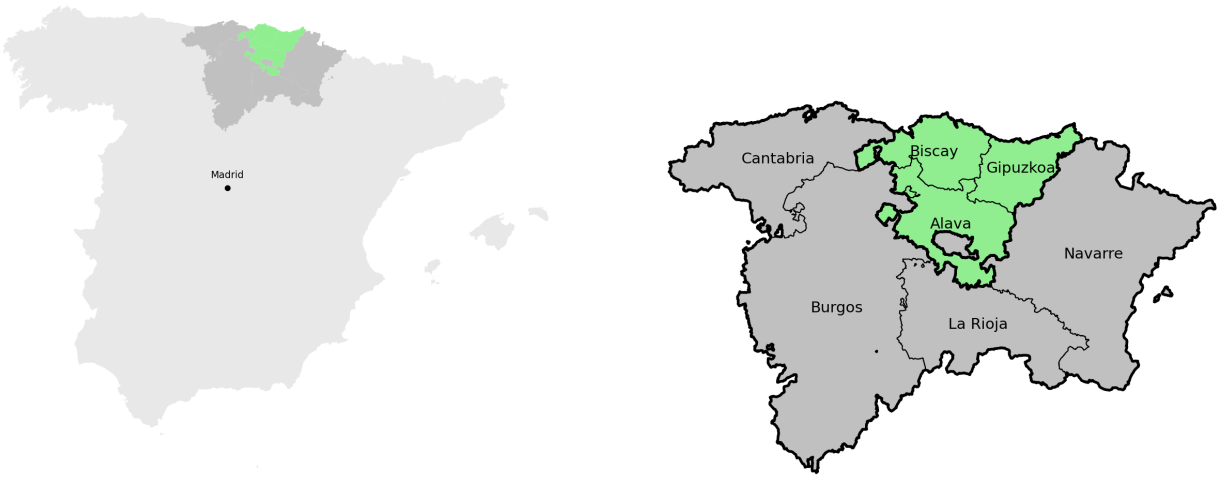
Alejandro Casado, Marco Giometti, Jose E. Gutierrez

David Martinez-Miera, Alexandra Matyunina, Tammaro Terracciano

A. Appendix: Institutional Details

Figure IA.I. Sample of Provinces

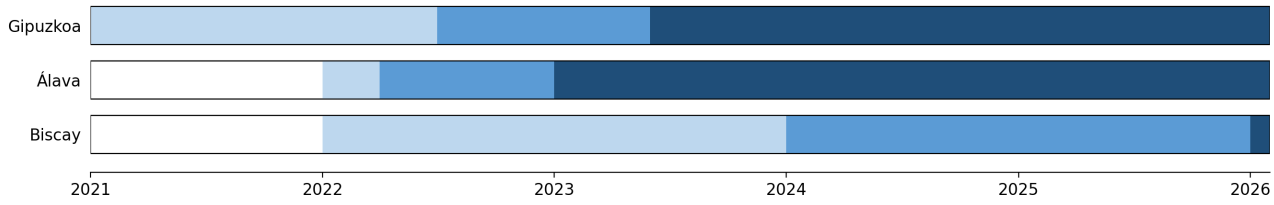
Panel IA.Ia shows the Spanish area we study, while, in panel IA.Ib, we zoom in on this green area. In this latter, the bold line delimits the Basque Country, and its provinces (Álava, Biscay, and Gipuzkoa). Neighboring provinces (in grey) are used as a control group in the empirical analysis.



(a) Spain **(b)** Basque Country and surroundings

Figure IA.II. Gantt Chart of the TicketBAI Staggered Introduction

White areas indicate no implementation; light blue areas indicate periods of voluntary adoption; blue areas indicate periods in which some industries face mandatory adoption while others remain voluntary; and dark blue areas indicate mandatory adoption for all industries.



B. Appendix: Data

B.1 Invoice-based Credit Definition

We define *invoice-based credit* as the set of credit operations reported to the Bank of Spain's Central Credit Register (CIR) under the heading *crédito comercial* (trade credit). Following the official Bank of Spain (BOE) reporting instructions, trade credit corresponds to credit granted on the basis of the purchase of collection rights (bills, invoices, or other documents) that arise from the deferral of payments associated with the sale of goods or the provision of services. Operationally, these credit lines encompass the discounting of commercial paper, factoring, and reverse factoring (payments to suppliers).

The BOE classifies these operations along two dimensions. The first is *recourse*: an operation is classified as *with recourse* (*con recurso*) when the assignor of the collection rights substantially retains the risks and rewards of the receivables, or when the reporting entity does not acquire control over their cash flows, so that the assignor is treated as the party bearing the direct credit risk; it is classified as *without recourse* (*sin recurso*) in the opposite case, and the obligor is then treated as the party bearing the direct credit risk. The second dimension applies within non-recourse operations and distinguishes *reverse factoring* (*pago a proveedores*), in which the reporting entity enters the operation through a contract with the obligor to pay its suppliers at maturity, from *factoring*, which covers all other non-recourse operations; each is further labeled as *with disbursement* (*con inversión*) when the entity may advance funds to the assignor before maturity, or *without disbursement* (*sin inversión*) when it is obligated to pay only on the maturity date. For all trade-credit operations, a credit limit is reported only when the entity has a committed line with the client.

Based on this taxonomy, the BOE categories that we aggregate under our definition of invoice-based credit are the following: discounting of commercial paper (*descuento de papel comercial*); other recourse operations (*resto de las operaciones*); reverse factoring with disbursement (*pago a proveedores con inversión*); reverse factoring without disbursement (*pago a proveedores sin inversión*); factoring with disbursement (*factoring con inversión*); factoring without disbursement (*factoring sin inversión*); and collection rights on regulated tariffs (*derechos de cobro sobre tarifas reguladas*).

B.2 Variable Definitions

Table IA.I. Variable definitions

Loan-level variables (Source: Central Credit Register (CIR), Bank of Spain)	
Loan Size	Outstanding used and unused committed credit granted under the loan.
Default	Indicator variable equal to 1 if the loan is currently classified as doubtful or past due 90 days or more.
One-year-ahead Realized Default	Indicator variable equal to 1 if the loan is classified as doubtful or past due 90 days or more over the next 4 quarters.
Default Rate	Share of the number of loans in default relative to the total number of loans outstanding.
Interest Rate	Interest rate charged on the loan.
Maturity	Loan maturity, measured in months.
Invoice-based Loan	Indicator variable equal to 1 if the loan is invoice-based (see Internet Appendix B.1).
ICO Loan	Indicator variable equal to 1 if the loan is secured by a public guarantee from the Spanish ICO.
Probability of Default (PD)	Probability of default of risk holders over a one-year period (Regulation (EU) No 575/2013) (bank-firm level information).
Bank Share in Firm Credit	Proportion of the total credit volume granted to a firm by a specific bank within a given period (bank-firm level information).
Credit-Type Controls:	(bank-firm or firm level information)
Share of Fixed-Rate Credit	Share of outstanding credit granted at a fixed interest rate.
Share of Term Loans	Share of outstanding credit granted through term loans.
Share of Credit Guaranteed by Physical Collateral	Share of outstanding credit guaranteed by physical collateral.
Weighted-Average Maturity	Credit-weighted average maturity of outstanding loans, measured in months.
Bank Credit over Total Assets	Outstanding committed bank credit divided by the firm's total assets.
Firm-level variables (Source: Mercantile Register, CIR and Financial Statements, Bank of Spain & Spanish Social Security Institute)	
Industry	One of the sectors in the 2-digit NACE 2009 industry classification
N. Loans	Number of outstanding loans granted to the firm.
N. Bank Relationships	Number of banks with which the firm has outstanding credit relationships.
N. IRB Bank Relationships	Number of IRB banks with which the firm has outstanding credit relationships.
Invoice Firm	Indicator variable equal to 1 if the firm has at least one outstanding invoice-based loan in 2021
Total Assets	Firm total assets as of year-end.
Size	Logarithm of total assets.
Sales	Firm sales as of year-end.
Leverage	Ratio of book total debt to total assets as of year-end.
Liquidity	Ratio of current assets to total assets as of year-end.
Tangibility	Ratio of tangible fixed assets to total assets as of year-end.
Profitability	Return on assets, calculated as EBIT divided by total assets.
Age	Firm age, measured as the difference between the current year and the year of incorporation.
Total Credit	Used and unused committed credit granted by Spanish banking institutions.
Invoice Credit	Used and unused committed credit described in Internet Appendix B.1 granted by Spanish banking institutions.
Non-Invoice Credit	Difference between total credit and invoice-based credit.
Share of ICO-guaranteed Credit	Share of the firm's total credit that is guaranteed by the Instituto de Crédito Oficial.
ERTE	Number of employees covered by the ERTE temporary employment adjustment scheme (province-industry-month level information).
New Relationship	Indicator equal to one if the firm originates a new loan with a bank not previously providing credit.
New Invoice Relationship	Indicator equal to one if the firm originates a new invoice-based loan with a bank not previously providing invoice credit.
New Non-Invoice Relationship	Indicator equal to one if the firm originates a new non-invoice loan with a bank not previously providing non-invoice credit.
Weighted Average Bank Controls:	(weights: each bank's share of the firm's credit)
Profitability	Return on assets, calculated as net income divided by total assets.
Liquidity	The ratio of cash holdings, cash balances with central banks, and other demand deposits to total assets.
Size	The logarithm of the bank's total assets.
Non-performing Loan Ratio	Non-performing loans divided by total assets.
Capitalization	Common Equity Tier 1 (CET1) capital divided by total assets.
Province-level variables (Source: CIR and Spanish Mercantile Register, Bank of Spain; Google)	
N. Firms	The number of firms with outstanding committed credit in the province.
Total Credit	Amount of total credit granted to firms in the province.
Share of SMEs	Share of firms in the province classified as small or medium-sized enterprises.
IRB Banks' Market Share	Share of total provincial corporate credit granted by IRB banks (calculated only on credit to SMEs).
Corporate Credit Market HHI	Herfindahl-Hirschman Index of the corporate credit market in the province (calculated only on credit to SMEs).
Interquartile Range	The difference between the 75th and 25th percentiles of the distribution of interest rates on newly originated loans or of probabilities of default for existing bank–firm relationships within a given quarter.
Standard Deviation	The standard deviation of the distribution of interest rates on newly originated loans or of probabilities of default for existing bank–firm relationships within a given quarter.
Share of Credit to Agriculture	Share of provincial corporate credit granted to SMEs in agriculture.
Share of Credit to Mining	Share of provincial corporate credit granted to SMEs in mining.
Share of Credit to Manufacturing	Share of provincial corporate credit granted to SMEs in manufacturing.
Share of Credit to Construction	Share of provincial corporate credit granted to SMEs in construction.
Share of Credit to Retail/Wholesale	Share of provincial corporate credit granted to SMEs in retail and wholesale trade.
Share of Credit to Transport	Share of provincial corporate credit granted to SMEs in transport.
Share of Credit to Hospitality	Share of provincial corporate credit granted to SMEs in hospitality.
Google Mobility index	Change in visitors to public transport hubs relative to a baseline from Google COVID-19 Mobility Reports.

B.3 Additional Summary Statistics

Table IA.II. Descriptive Statistics: Invoice vs. Non-Invoice Firms

This table presents firm-level descriptive statistics for invoice firms versus non-invoice firms. All variables are measured as of the end of 2021, the pre-reform period, and are defined in Table IA.I. Column (1) reports the mean for invoice firms, Column (2) reports the mean for non-invoice firms, Column (3) reports the difference in means between invoice and non-invoice firms, and Column (4) reports the p-value of a two-sided test for the equality of means across the two groups. Standard deviations appear in parentheses, except for N, for which the relative share appears in parentheses. For the tests of equality of means, standard errors are clustered at the province×industry level. p-values are reported to two decimal digits, with values below 0.01 reported as < 0.01. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice Firms	Non-Invoice Firms	Diff.	P-value
	(1)	(2)	(3)	(4)
N	5,190 (23.34%)	17,049 (76.66%)		
N. loans	14.81 (18.84)	4.25 (7.06)	10.56	$< 0.01^{***}$
N. bank relationships	3.87 (2.53)	1.76 (1.13)	2.11	$< 0.01^{***}$
N. IRB bank relationships	1.56 (1.11)	0.75 (0.74)	0.81	$< 0.01^{***}$
Total assets, million €	2.92 (5.01)	1.21 (2.85)	1.71	$< 0.01^{***}$
Sales, million €	3.35 (5.71)	0.82 (1.98)	2.53	$< 0.01^{***}$
Leverage	0.35 (0.24)	0.31 (0.31)	0.04	$< 0.01^{***}$
Liquidity	0.71 (0.23)	0.61 (0.31)	0.10	$< 0.01^{***}$
Tangibility	0.23 (0.21)	0.27 (0.29)	–0.05	$< 0.01^{***}$
Profitability	0.02 (0.14)	0.03 (0.19)	–0.01	$< 0.01^{***}$
Age	20.83 (12.09)	16.28 (10.79)	4.54	$< 0.01^{***}$
Share of ICO-guaranteed credit	0.41 (0.30)	0.30 (0.38)	0.11	$< 0.01^{***}$

Table IA.III. Descriptive Statistics: Invoice vs. Non-Invoice Firms, Treated vs. Control

This table presents firm-level descriptive statistics for invoice firms (Panel A) versus non-invoice firms (Panel B), split by treated and control groups. All variables are measured as of the end of 2021, the pre-reform period, and are defined in Table IA.I. Within each panel, Column (1) reports the mean for the treated group and Column (2) the mean for the control group. Column (3) reports the difference in means between treated and control firms, and Column (4) reports the p-value of a two-sided test for the equality of means across the two groups. Standard deviations appear in parentheses, except for N, for which the relative share appears in parentheses. For the tests of equality of means, standard errors are clustered at the province×industry level. p-values are reported to two decimal digits. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Treated	Control	Diff.	P-value
	(1)	(2)	(3)	(4)
Panel A: Invoice Firms				
N	2,252 (43.39%)	2,938 (56.61%)		
N. loans	14.37 (18.77)	15.15 (18.88)	–0.78	0.31
N. bank relationships	3.78 (2.51)	3.93 (2.54)	–0.14	0.15
N. IRB bank relationships	1.51 (1.14)	1.60 (1.10)	–0.09	0.13
Total assets, million €	2.72 (4.77)	3.07 (5.17)	–0.35	0.40
Sales, million €	3.07 (5.39)	3.56 (5.94)	–0.49	0.19
Leverage	0.35 (0.26)	0.35 (0.23)	0.00	0.70
Liquidity	0.74 (0.23)	0.69 (0.23)	0.05	0.05*
Tangibility	0.20 (0.21)	0.25 (0.21)	–0.05	0.04**
Profitability	0.01 (0.15)	0.02 (0.12)	–0.01	0.05*
Age	20.14 (11.59)	21.34 (12.42)	–1.20	0.31
Share of ICO-guaranteed credit	0.39 (0.30)	0.43 (0.30)	–0.05	0.02**
Panel B: Non-Invoice Firms				
N	7,545 (44.25%)	9,504 (55.75%)		
N. loans	4.25 (8.92)	4.24 (5.11)	0.01	0.98
N. bank relationships	1.73 (1.13)	1.78 (1.13)	–0.04	0.42
N. IRB bank relationships	0.72 (0.75)	0.77 (0.73)	–0.05	0.35
Total assets, million €	1.27 (3.08)	1.17 (2.66)	0.10	0.59
Sales, million €	0.89 (2.25)	0.77 (1.74)	0.12	0.47
Leverage	0.29 (0.31)	0.33 (0.31)	–0.03	0.16
Liquidity	0.64 (0.32)	0.59 (0.31)	0.04	0.42
Tangibility	0.24 (0.28)	0.30 (0.28)	–0.06	0.05*
Profitability	0.03 (0.20)	0.03 (0.18)	0.00	0.48
Age	16.06 (10.54)	16.46 (10.99)	–0.39	0.56
Share of ICO-guaranteed credit	0.26 (0.36)	0.34 (0.39)	–0.08	0.02**

Table IA.IV. Summary Statistics: Full Sample

This table presents summary statistics for the full sample period, January 2021 to December 2024. Panel A reports statistics at the bank-firm level, Panel B at the firm level, and Panel C at the province level. Panel A reports all bank-firm-month observations and Panel B all firm-month observations for our main sample. Panel C reports observations at the province-month level. All variables are defined in Table IA.I. Column (1) reports the number of observations, Column (2) the mean, Column (3) the standard deviation, Column (4) the minimum, Columns (5)-(7) the 25th, 50th (median), and 75th percentiles, and Column (8) the maximum. Credit variables are expressed in thousands of euros. Shares, ratios, and probabilities are expressed as fractions, except where otherwise noted.

	N	Mean	SD	Min	25%	Median	75%	Max
Panel A: Bank-Firm Level								
Total credit, thousand €	1,899,631	202.01	363.82	0.78	18.55	69.22	207.65	2,298.30
Invoice credit, thousand €	1,899,631	16.16	61.46	0.00	0.00	0.00	0.00	438.11
Non-Invoice credit, thousand €	1,899,631	181.18	327.16	0.00	15.30	61.18	190.00	2,061.24
Probability of default, %	896,297	5.95	18.06	0.03	0.31	1.02	2.77	100.00
Interest rate, %	1,575,299	3.73	3.27	0.60	1.76	2.88	4.74	23.80
Maturity, months	1,520,548	81.26	52.51	1.50	54.10	63.36	97.40	305.30
Share of invoice-based credit	1,899,631	0.06	0.19	0.00	0.00	0.00	0.00	1.00
Share of ICO-guaranteed credit	1,899,631	0.31	0.40	0.00	0.00	0.00	0.70	1.00
Panel B: Firm Level								
Total credit, thousand €	971,305	398.40	910.79	1.50	23.85	90.31	312.69	6,055.21
Invoice credit, thousand €	971,305	28.76	121.29	0.00	0.00	0.00	0.00	916.03
Non-Invoice credit, thousand €	971,305	359.73	801.66	1.20	22.72	85.73	294.40	5,311.05
Interest rate, %	827,532	3.71	3.29	0.65	1.87	2.93	4.54	23.80
Interest rate, invoice credit, %	73,842	4.43	2.33	0.70	2.51	4.32	5.95	11.50
Interest rate, non-invoice credit, %	825,621	3.70	3.30	0.65	1.86	2.90	4.51	23.80
Share of invoice firms	971,305	0.24	0.43	0.00	0.00	0.00	0.00	1.00
Age	971,305	17.74	11.38	0.00	8.00	17.00	25.00	121.00
Leverage	971,305	0.31	0.29	0.00	0.07	0.24	0.47	1.52
Liquidity	971,305	0.64	0.30	0.01	0.41	0.70	0.90	1.00
Tangibility	971,305	0.26	0.27	0.00	0.04	0.17	0.42	0.97
Profitability	971,305	0.02	0.17	–0.79	–0.01	0.02	0.08	0.52
ERTE	971,305	74.02	401.77	0.00	0.00	5.00	21.00	10,246.00
New relationship, % (quarterly)	322,591	2.26	14.87	0.00	0.00	0.00	0.00	100.00
New invoice relationship, % (quarterly)	322,591	1.24	11.07	0.00	0.00	0.00	0.00	100.00
New non-invoice relationship, % (quarterly)	322,591	2.05	14.15	0.00	0.00	0.00	0.00	100.00
Panel C: Province Level								
Share of SMEs	288	0.95	0.01	0.92	0.94	0.96	0.96	0.97
IRB banks' market share	288	0.50	0.06	0.39	0.45	0.50	0.54	0.62
Corporate credit market HHI	288	0.13	0.03	0.09	0.11	0.12	0.16	0.17
Share of credit to:								
Agriculture	288	0.02	0.01	0.01	0.02	0.02	0.02	0.04
Mining	288	0.003	0.00	0.00	0.00	0.00	0.00	0.01
Manufacturing	288	0.31	0.07	0.20	0.24	0.32	0.37	0.42
Construction	288	0.12	0.02	0.08	0.10	0.11	0.14	0.16
Retail/Wholesale	288	0.25	0.03	0.20	0.22	0.25	0.27	0.30
Transport	288	0.06	0.03	0.02	0.04	0.06	0.08	0.12
Hospitality	288	0.05	0.01	0.04	0.04	0.05	0.06	0.08
Google Mobility Index	288	–0.80	17.45	–47.87	–6.17	0.00	0.00	120.74

C. Appendix: Credit Market Dynamics

In this appendix, we report supplementary analyses and robustness checks that complement the credit-market evidence in Section 4. Throughout, we focus on estimates at the bank-firm-month level for the split between invoice and non-invoice firms, except when indicated otherwise. The goal is to assess whether our main findings are robust to alternative samples, geographic restrictions, fixed-effects specifications, and other potential confounding factors, and to provide additional evidence on firm heterogeneity. Taken together, the results support the robustness of the credit-market effects documented in the main text.

C.1 Credit Scaled by Revenues

To further address the concern that our results reflect newly reported sales rather than changes in the informational properties of invoices, we re-estimate our baseline specifications at both the bank-firm-month and firm-month level using invoice-based credit scaled by firm revenues as the dependent variable. If eInvoicing mainly increased the stock of reported (and thus pledgeable) receivables, invoice credit should rise roughly proportionally with reported revenues, leaving the ratio unchanged. Table IA.V reports the results for the full sample and for the subsample of invoice firms. In both cases, invoice credit increases relative to revenues after the reform, confirming that the expansion of invoice-based borrowing is not driven only by newly reported sales.

Table IA.V. Robustness: Invoice-Based Credit over Revenues

This table reports estimates of ω from regressions where the dependent variable is invoice-based credit divided by current firm revenues, over the main sample from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. Column 1 reports estimates for the full sample; column 2 restricts the sample to invoice firms. In Panel A, the unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \omega \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

In Panel B, the unit of observation is a firm-month, invoice-based credit is aggregated across all banks lending to firm f in month t, and we estimate:

$$\mathbb{E}[Y_{f,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_f + \omega \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_{\bar{b}} \cdot \bar{X}_{b,f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

In Panel A, $Y_{f,b,t}$ is invoice-based credit extended by bank b to firm f in month t, divided by firm revenues. In Panel B, $Y_{f,t}$ is firm-level invoice-based credit divided by firm revenues. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. Panel A includes industry×size-tercile×month, firm×bank, and bank×month fixed effects. Panel B includes industry×size-tercile×month and firm fixed effects. In both panels, $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, and age. In Panel A, this vector also includes bank-firm-level controls: the share of bank-firm credit guaranteed by ICO and an indicator for whether the firm is in default with that bank. In Panel B, these two variables are replaced by their firm-level counterparts: the share of the firm's total credit guaranteed by ICO and an indicator for whether the firm is in default with any bank. Average bank controls in Panel B are denoted by $\bar{X}_{b,f,t-1}$ and are computed as firm-level weighted averages, with weights based on each bank's share of the firm's credit. These controls include bank profitability, liquidity, size, non-performing loan ratio, and capitalization. In both panels, $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. The row *Controls* indicates that all specifications include firm and Covid controls in Panel A, and firm, Covid, and average bank controls in Panel B. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice-Based Credit/Revenues	
	Full Sample	Invoice Firms
	(1)	(2)
Panel A: Bank-firm level		
Post×Treated	0.0258***	0.0211**
	(0.0085)	(0.0086)
Industry×Size×Month FE	Yes	Yes
Firm×Bank FE	Yes	Yes
Bank×Month FE	Yes	Yes
Controls	Yes	Yes
Obs.	378,497	349,939
Pseudo R-Squared	0.15	0.15
Panel B: Firm level		
Post×Treated	0.0324***	0.0383***
	(0.0106)	(0.0108)
Industry×Size×Month FE	Yes	Yes
Firm FE	Yes	Yes
Controls	Yes	Yes
Obs.	197,784	166,870
Pseudo R-Squared	0.19	0.17

C.2 Subsample of Firms with PDs

Because the information-quality analysis in Section 5.2 is necessarily restricted to firms for which at least one IRB bank reports a usable PD, we re-estimate the main credit specifications on this subsample to confirm that our results hold. Table IA.VI shows that the results are very

similar to the baseline ones: invoice firms experience an increase in invoice credit and total credit, whereas non-invoice firms experience a decline in total credit.

Table IA.VI. Robustness: Firms Borrowing from IRB Banks

This table reports estimates of β over the subsample of firms for which at least one IRB bank reports a usable PD, from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. The unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

$Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. For invoice firms, the dependent variable is invoice-based credit in column 1, non-invoice credit in column 2, and total credit in column 3. For non-invoice firms, the dependent variable is total credit in column 4. We cannot estimate specifications with invoice-based credit as the dependent variable for non-invoice firms because, by construction, these firms have no positive invoice-based credit in the pre-reform period. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. The specification includes industry×size-tercile×month, firm×bank, and bank×month fixed effects. $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, age, the share of bank-firm credit guaranteed by ICO, and an indicator for whether the firm is in default with that bank. $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. The row *Controls* indicates that all specifications include firm and Covid controls. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice Firms			**Non-Invoice Firms**
	Invoice	Non-Invoice	Total	Total
	(1)	(2)	(3)	(4)
Post×Treated	0.0438***	0.0012	0.0056*	–0.0061**
	(0.0080)	(0.0032)	(0.0031)	(0.0026)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm×Bank FE	Yes	Yes	Yes	Yes
Bank×Month FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	439,138	631,504	654,230	697,760
Pseudo R-Squared	0.75	0.93	0.93	0.96

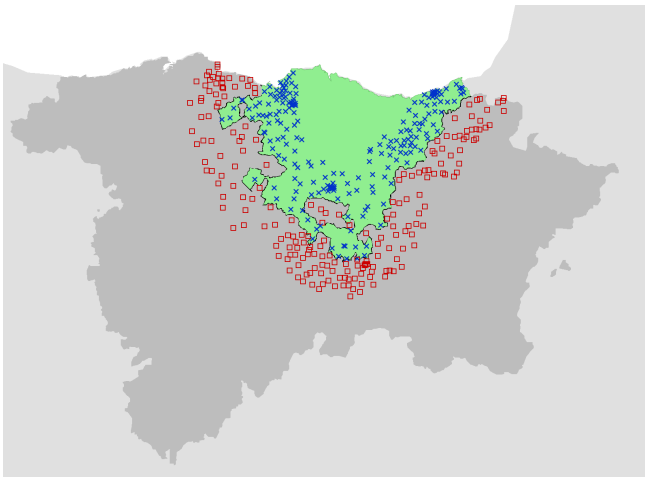
C.3 Subsample of Firms near the Basque Country Border

To further mitigate concerns about unobserved heterogeneity between treated and control provinces, we restrict the comparison to firms located near the Basque Country border. The underlying idea is that firms on either side of the border are more likely to face similar local economic conditions, so that comparing nearby treated and control firms provides a more demanding test of the reform's effects. We therefore restrict the sample ex ante to zip codes within a certain distance (either 20, 30, or 50 km) from the Basque Country border and re-estimate the baseline bank-firm specifications.[42] Figure IA.III illustrates the geographic distribution of the resulting samples, and Table IA.VII reports the estimates. The paper's main findings remain unchanged.

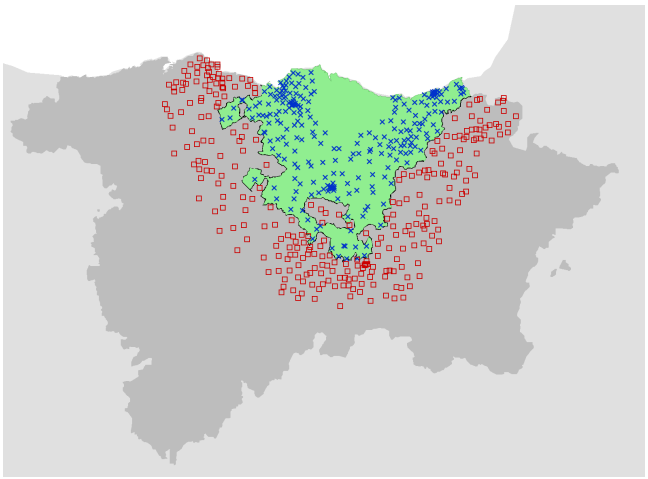
42. For consistency with the baseline design, we exclude Gipuzkoa from this exercise.

Figure IA.III. Geographic Distribution of Firms by Distance from the Basque Country Land Border

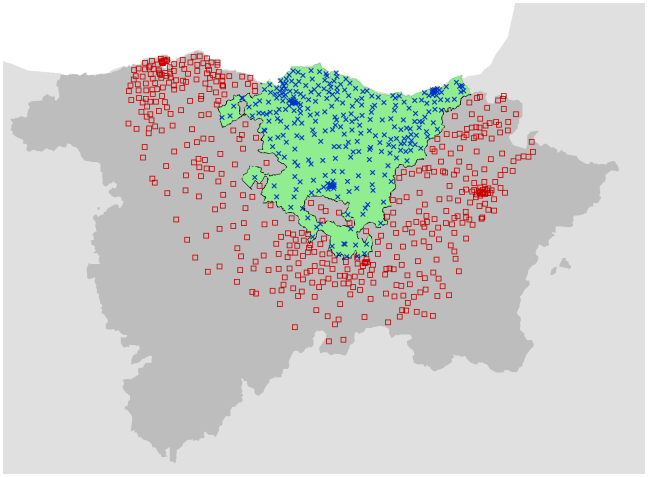
The green area is the Basque Country (treatment); the medium-grey area covers the four adjacent control provinces (Cantabria, Burgos, La Rioja, and Navarre). Blue Xs mark firms located inside the Basque Country; red squares mark firms in the control provinces. Panels (a)–(c) show the subsample of firms whose zip code centroid lies within 20, 30, and 50 km of the Basque Country's land border, respectively.



(a) Within 20 km of the Basque Country land border



(b) Within 30 km of the Basque Country land border



(c) Within 50 km of the Basque Country land border

Table IA.VII. Robustness: Firms Near the Basque Country Border

This table reports estimates of β from regressions estimated separately for invoice firms and non-invoice firms located within a given distance from the land border of the Basque Country, from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. Each panel corresponds to a different distance bandwidth. The unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

$Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. For invoice firms, the dependent variable is invoice-based credit in column 1, non-invoice credit in column 2, and total credit in column 3. For non-invoice firms, the dependent variable is total credit in column 4. We cannot estimate specifications with invoice-based credit as the dependent variable for non-invoice firms because, by construction, these firms have no positive invoice-based credit in the pre-reform period. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. The specification includes industry×size-tercile×month, firm×bank, and bank×month fixed effects. $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, age, the share of bank-firm credit guaranteed by ICO, and an indicator for whether the firm is in default with that bank. $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. The row *Controls* indicates that all specifications in every panel include firm and Covid controls. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice Firms			**Non-Invoice Firms**
	Invoice	Non-Invoice	Total	Total
	(1)	(2)	(3)	(4)
Panel A: Bandwidth = 20 km				
Post×Treated	0.0319**	0.0196***	0.0148***	–0.0201***
	(0.0141)	(0.0048)	(0.0045)	(0.0033)
Obs.	195,794	280,983	291,895	514,192
Pseudo R-Squared	0.75	0.94	0.93	0.97
Panel B: Bandwidth = 30 km				
Post×Treated	0.0683***	0.0129***	0.0154***	–0.0258***
	(0.0143)	(0.0043)	(0.0037)	(0.0031)
Obs.	254,571	363,373	377,032	622,044
Pseudo R-Squared	0.75	0.94	0.93	0.97
Panel C: Bandwidth = 50 km				
Post×Treated	0.0535***	0.0079**	0.0123***	–0.0111***
	(0.0078)	(0.0031)	(0.0029)	(0.0026)
Obs.	406,833	578,372	599,523	951,417
Pseudo R-Squared	0.75	0.94	0.93	0.97
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm×Bank FE	Yes	Yes	Yes	Yes
Bank×Month FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

C.4 Leave-One-Out Province Analysis

Because our identification relies on a small number of treated and control provinces, one may worry that idiosyncratic dynamics in a single province drive our results. To assess this, we re-estimate the baseline specification while sequentially excluding each province from the sample. As a complementary check, we also report estimates from a staggered design that includes

Gipuzkoa. We exclude this province from the baseline analysis because its voluntary phase begins in 2021, leaving no reliable pre-reform period in our sample (see Section 2). However, the staggered design allows treatment timing to vary across provinces according to the start of TicketBAI in each province. Table IA.VIII shows that the main patterns remain overall stable across these alternative samples. This indicates that the documented effects are not meaningfully driven by any single province, nor by the exclusion of Gipuzkoa from the baseline design.

Table IA.VIII. Robustness: Leave-One-Out Estimates and Gipuzkoa Inclusion

This table reports estimates of β from regressions estimated separately for invoice firms and non-invoice firms, using Poisson Pseudo-Maximum Likelihood. Panel A re-estimates the baseline specification, excluding one province at a time from the main sample from January 2021 to December 2024. In these specifications, treatment starts in January 2022 for the treated provinces (Álava and Biscay). Panel B includes Gipuzkoa in the sample and estimates a staggered difference-in-differences specification in which treatment starts in each treated province when its voluntary TicketBAI period begins. The unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot \mathit{Treatment}_{p,t} + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

In Panel A, $\mathit{Treatment}_{p,t}$ equals $\mathit{Post}_t \times \mathit{Treat}_p$, where Post_t is an indicator equal to one from January 2022 onward and Treat_p is an indicator equal to one for firms headquartered in treated provinces. In Panel B, $\mathit{Treatment}_{p,t}$ is an indicator equal to one once province p enters the voluntary TicketBAI period. $Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. For invoice firms, the dependent variable is invoice-based credit in column 1, non-invoice credit in column 2, and total credit in column 3. For non-invoice firms, the dependent variable is total credit in column 4. We cannot estimate specifications with invoice-based credit as the dependent variable for non-invoice firms because, by construction, these firms have no positive invoice-based credit in the pre-reform period. The specification includes industry×size-tercile×month, firm×bank, and bank×month fixed effects. $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, age, the share of bank-firm credit guaranteed by ICO, and an indicator for whether the firm is in default with that bank. $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice Firms			**Non-Invoice Firms**
	Invoice	Non-Invoice	Total	Total
	(1)	(2)	(3)	(4)
Panel A: Leave-One-Out Estimates (common treatment date)				
Excluding Álava	0.0529***	–0.0104***	–0.0021	–0.0082***
	(0.0079)	(0.0027)	(0.0030)	(0.0026)
Excluding Biscay	0.0248**	0.0245***	0.0209***	–0.0048*
	(0.0098)	(0.0037)	(0.0039)	(0.0028)
Excluding Burgos	0.0407***	0.0068**	0.0093***	–0.0066***
	(0.0078)	(0.0032)	(0.0031)	(0.0023)
Excluding Cantabria	0.0420***	0.0080**	0.0114***	–0.0083***
	(0.0082)	(0.0032)	(0.0032)	(0.0022)
Excluding Navarre	0.0247***	–0.0051	–0.0036	–0.0067***
	(0.0076)	(0.0032)	(0.0028)	(0.0022)
Excluding La Rioja	0.0468***	0.0023	0.0071**	–0.0041*
	(0.0072)	(0.0033)	(0.0032)	(0.0022)
Panel B: Including Gipuzkoa (province-specific treatment dates)				
Including Gipuzkoa	0.0276***	0.0089***	0.0096***	0.0011
	(0.0069)	(0.0029)	(0.0025)	(0.0021)

C.5 Accounting for Bank-Specific Sector Shocks

A potential concern is that the baseline fixed-effects structure may not fully absorb time-varying credit-supply shocks at the bank-sector level. To address this, we re-estimate the baseline specification using a more demanding fixed-effects combination that includes bank×industry×month variation. Table IA.IX shows that the estimated effects remain very similar to the baseline ones. This indicates that the main results are unlikely to be driven by bank-specific shifts in lending to particular sectors that may correlate with treatment exposure.

Table IA.IX. Robustness: Bank-Industry-Month Fixed Effects

This table reports estimates of β from regressions estimated separately for invoice firms and non-invoice firms over our main sample from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. The unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,i,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

$Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. For invoice firms, the dependent variable is invoice-based credit in column 1, non-invoice credit in column 2, and total credit in column 3. For non-invoice firms, the dependent variable is total credit in column 4. We cannot estimate specifications with invoice-based credit as the dependent variable for non-invoice firms because, by construction, these firms have no positive invoice-based credit in the pre-reform period. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. The specification includes industry×size-tercile×month, firm×bank, and bank×industry×month fixed effects. $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, age, the share of bank-firm credit guaranteed by ICO, and an indicator for whether the firm is in default with that bank. $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. The row *Controls* indicates that all specifications include firm and Covid controls. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice Firms			**Non-Invoice Firms**
	Invoice	Non-Invoice	Total	Total
	(1)	(2)	(3)	(4)
Post×Treated	0.0433***	0.0035	0.0070**	–0.0096***
	(0.0072)	(0.0031)	(0.0031)	(0.0021)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm×Bank FE	Yes	Yes	Yes	Yes
Bank×Industry×Month FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	487,736	705,648	731,248	1,150,401
Pseudo R-Squared	0.77	0.94	0.94	0.97

C.6 SII: *Suministro Inmediato de Información*

Another potential concern is that our results may be confounded by the *Suministro Inmediato de Información* (SII), a Spanish VAT reporting system that requires certain firms to report billing data to the tax authority within 4 days.[43] We view this as unlikely for three reasons.

43. Specifically, the eligible entities are: large businesses (i.e., firms with turnover above €6 million); VAT groups; firms registered with REDEME (monthly VAT return registry); holders of tax warehouses for gasoline, diesel or bio-

First, the SII entered into force in 2017, well before our sample starts, and applies in both treated and control provinces. Second, our fixed-effects structure absorbs persistent firm-level differences related to SII eligibility. Third, Table IA.X shows that our main results remain unchanged when we exclude firms most likely affected by the SII, namely those with turnover above €6 million and those in fuel warehousing-related industries.[44] These results indicate that the documented credit effects are unlikely to be driven by the SII.

Table IA.X. Robustness: Excluding Firms Potentially Subject to SII

This table reports estimates of β from regressions estimated separately for invoice firms and non-invoice firms over a restricted sample from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. The sample excludes firms with turnover above €6 million and firms in 2-digit NACE-2009 industries 19, 46, 47, and 52, which are potentially subject to the *Suministro Inmediato de Información* regime. The unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\}.$$

$Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. For invoice firms, the dependent variable is invoice-based credit in column 1, non-invoice credit in column 2, and total credit in column 3. For non-invoice firms, the dependent variable is total credit in column 4. We cannot estimate specifications with invoice-based credit as the dependent variable for non-invoice firms because, by construction, these firms have no positive invoice-based credit in the pre-reform period. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. The specification includes industry×size-tercile×month, firm×bank, and bank×month fixed effects. $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, age, the share of bank-firm credit guaranteed by ICO, and an indicator for whether the firm is in default with that bank. $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. The row *Controls* indicates that all specifications include firm and Covid controls. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice Firms			**Non-Invoice Firms**
	Invoice	Non-Invoice	Total	Total
	(1)	(2)	(3)	(4)
Post×Treated	0.0355***	0.0002	0.0059**	–0.0041***
	(0.0115)	(0.0026)	(0.0023)	(0.0016)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm×Bank FE	Yes	Yes	Yes	Yes
Bank×Month FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	192,861	296,758	304,969	641,175
Pseudo R-Squared	0.76	0.96	0.95	0.98

C.7 Alternative Clustering of Standard Errors

We cluster standard errors at the province-time level in our main specifications (Equation (1)). This choice reflects three features of our design. First, while the eInvoicing reform

fuels included in the objective scope of the Hydrocarbon Tax; and entrepreneurs or professionals who extract these products from tax warehouses (from 01/01/2025). Additional information about the SII can be found at https://sede.agenciatributaria.gob.es/Sede/en_gb/iva/suministro-inmediato-informacion/informacion-general.html.

44. The law does not explicitly mention specific NACE sectors; we assign them manually. To be conservative, we discard the following 2-digit NACE codes: 19 - Manufacture of coke and refined petroleum products; 46 - Wholesale trade, except for motor vehicles and motorcycles; 47 - Retail trade, except for motor vehicles and motorcycles; 52 — Warehousing and support activities for transportation.

was introduced at the province level, its implementation varied over time across voluntary and mandatory phases, so that the relevant policy variation in our estimating equations is defined at the province-month level. Second, our preferred specifications include a rich set of fixed effects—industry×size-tercile×month, bank×month, and firm×bank—which absorb the main channels through which province-level or aggregate shocks could affect credit outcomes. The industry×size-tercile×month effects in particular control for industry-time credit demand, while also absorbing size-specific shocks, including any potential differential exposure defined at the firm-size level that overlaps with our sample period. Third, we report a more demanding triple-interaction specification with province×month×industry×size fixed effects (Table 8), which absorbs shocks at the most granular implementation-exposure cell. After this conditioning, identification comes from residual variation between invoice and non-invoice firms within the same province-time-industry-size, alleviating concerns about residuals' autocorrelation. This inference approach is consistent with the design-based perspective of Abadie et al. (2023), who emphasize that clustering choices should be guided by the sampling and treatment-assignment process rather than by residual correlation alone.

Nevertheless, to address possible remaining concerns, we discuss alternative clustering choices and report their p-values in Table IA.XI. In the first alternative, we use heteroskedasticity-robust standard errors, which allow us to be agnostic about the level of the treatment and possible autocorrelation in the errors. The significance of these estimates is consistent with the baseline results in Table 4.

Next, we cluster standard errors at the industry-time and province-industry-time levels, since reform implementation timing depended on the industry (see Section 2). The caveat is that our industry variable is the standard NACE classification, while TicketBAI uses IAE, for which there is no direct mapping. Nonetheless, also in these cases, our estimates are consistent with the baseline results in Table 4.

Finally, we focus on the province dimension alone. The issue with this approach is that the conventional cluster-robust inference is not reliable, because the design contains only six provinces, two of which are treated. Our baseline province-month clustering matches the observed policy-exposure cell, but it does not by itself address arbitrary serial correlation in province-level shocks. We therefore treat province-level inference as the relevant conservative benchmark. MacKinnon and Webb (2020) show that, with very few treated clusters, conventional cluster-robust t-tests can severely over-reject; the usual restricted wild cluster bootstrap can severely

under-reject, while unrestricted variants can over-reject. For PPML, residual-based additive wild-bootstrap pseudo-outcomes are also not straightforward because they may be negative, and score/multiplier variants do not eliminate the finite-sample concern created by having 6 provinces, of which only 2 are treated. We therefore report exact province-level randomization inference over all $\binom{6}{2} = 15$ two-province placebo assignments, using the cluster-robust t-statistic as the test statistic, in line with the recommendation of MacKinnon and Webb (2020).

It is important to underscore how stringent this procedure is in our setting. With only 15 admissible treatment-pair assignments, the exact two-sided p-value is necessarily a multiple of $1/15$, so the smallest value the test can ever produce is $1/15 \approx 0.067$. As a mechanical consequence, statistical significance at the conventional 5% level is *unattainable* by design, regardless of the magnitude or precision of the underlying effect; the most decisive verdict the procedure can return is significance at the 10% level, and only when the actual Álava–Biscay assignment yields a more extreme $|t|$ than *all fourteen* alternative two-province placebos. The procedure is therefore heavily skewed toward non-rejection: it requires the realized treatment to produce a uniformly more extreme test statistic than every other province pair, including pairs that mix one treated and one control province and that are mechanically correlated with the actual treatment. Failure to reject under this test is consequently weak evidence of a null, while a rejection – even at 10% – constitutes unusually strong evidence in favor of an effect. Against this backdrop, the realized assignment delivers RI-$t = 1/15 \approx 0.067$ for invoice-based credit, the minimum attainable exact p-value in this design. We interpret this as the strongest evidence that the randomization-inference procedure can provide for the existence of a true effect, while acknowledging that the six-province design necessarily limits the resolution of any inference procedure built on the province-level permutations.

Table IA.XI. eInvoicing and Credit Supply: Inference Robustness

This table reports the coefficient on Post×Treated from the regression specified in Equation (1), separately for invoice firms (columns 1–3) and non-invoice firms (column 4). All specifications are estimated using Poisson-Pseudo Maximum Likelihood at the bank-firm-month level and include Industry × Size × Month, Firm × Bank, and Bank × Month fixed effects, firm controls, and Covid controls. The first row reports the point estimate, mechanically identical across inference choices. Subsequent rows report two-sided p-values (in parentheses) under alternative variance–covariance assumptions: heteroskedasticity-robust (unclustered) inference; cluster-robust standard errors at the indicated level (using the normal-distribution reference); and randomization inference at the province level following MacKinnon and Webb (2020), enumerating all $\binom{6}{2} = 15$ placebo treatment-pair assignments and computing $p_t^* = \#\{|\hat{t}_r^*| \geq |\hat{t}|\}/15$ from the cluster-robust t-statistic. The smallest p-value the randomization-inference procedure can deliver in this design is $1/15 \approx 0.0667$. The row *Controls* indicates that all specifications include firm and Covid controls.

	Invoice Firms			**Non-Invoice Firms**
	Invoice	Non-Invoice	Total	Total
	(1)	(2)	(3)	(4)
Post×Treated	0.0407	0.0028	0.0064	–0.0070
Standard Errors (p-values are reported):				
Heteroskedasticity Robust	(0.0000)	(0.3185)	(0.0207)	(0.0085)
Clustered at Industry × Month Level	(0.0000)	(0.3664)	(0.0848)	(0.0017)
Clustered at Province × Industry × Month Level	(0.0000)	(0.3286)	(0.0405)	(0.0011)
Clustered at Province Level (with only 6 clusters):				
Conventional Approach	(0.0103)	(0.8389)	(0.5800)	(0.0837)
Randomization-inference Approach (RI-t)	(0.0667)	(0.9333)	(0.5333)	(0.2667)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm×Bank FE	Yes	Yes	Yes	Yes
Bank×Month FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	499,095	714,300	739,956	1,159,022
Pseudo R-Squared	0.75	0.93	0.93	0.97

C.8 Additional Firm Heterogeneity

We also explore whether the effects of eInvoicing vary across firms with different pre-reform characteristics. To do so, we split the sample along several firm-level dimensions (size, age, tangibility, and receivables intensity) measured before the reform, and re-estimate the baseline specification within each subsample. Table IA.XII reports the results and complements the main heterogeneity analysis in the paper, which focuses on the distinction between invoice and non-invoice firms.

In Panel A, invoice credit increases for both size groups, but the effect is substantially larger for below-median-size firms, which also experience a statistically significant decline in non-invoice and total credit; the corresponding estimates for above-median firms are close to zero and statistically insignificant. Panel B shows a similar pattern by age: the increase in invoice credit

is concentrated among below-median-age firms, which also experience a decline in non-invoice and total credit, while above-median-age firms display small positive effects on non-invoice and total credit and no significant effect on invoice credit. The same holds in Panel C for below-median-tangibility firms, although the decline in their total credit is not statistically significant, with mostly insignificant estimates for above-median-tangibility firms. Finally, in Panel D, invoice credit increases for firms with above-median receivables over sales, with no significant effects on their non-invoice or total credit, and no significant effects for below-median firms.

Overall, we observe a positive invoice-based credit response among smaller, younger, low-tangibility, and high-receivables firms, and this increase is accompanied by unchanged or lower total credit. These results support a reallocation toward invoice-based credit among firms for which receivable information is most relevant.

Table IA.XII. Additional Heterogeneity by Firm Characteristics

This table reports estimates of β from regressions estimated separately by firm characteristics over the main sample from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. Each panel splits firms at the median of a pre-reform firm characteristic: size in Panel A, age in Panel B, tangibility in Panel C, and receivables over sales in Panel D. Within each panel, columns 1–3 report estimates for firms above the median, while columns 4–6 report estimates for firms below the median. The unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

$Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. The dependent variable is invoice-based credit in columns 1 and 4, non-invoice credit in columns 2 and 5, and total credit in columns 3 and 6. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. The specification includes industry×size-tercile×month, firm×bank, and bank×month fixed effects. $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, age, the share of bank-firm credit guaranteed by ICO, and an indicator for whether the firm is in default with that bank. $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Above the Median			**Below the Median**		
	Invoice	Non-Invoice	Total	Invoice	Non-Invoice	Total
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Size						
Post×Treated	0.0433***	0.0018	0.0020	0.1506***	–0.0174***	–0.0112***
	(0.0077)	(0.0020)	(0.0019)	(0.0219)	(0.0034)	(0.0033)
Obs.	456,985	1,238,667	1,261,769	94,564	633,734	637,393
Pseudo R-Squared	0.74	0.94	0.94	0.74	0.94	0.94
Panel B: Age						
Post×Treated	0.0079	0.0077***	0.0061***	0.1018***	–0.0157***	–0.0100***
	(0.0105)	(0.0023)	(0.0024)	(0.0125)	(0.0033)	(0.0029)
Obs.	347,358	1,001,149	1,020,348	202,540	870,499	877,994
Pseudo R-Squared	0.76	0.95	0.95	0.76	0.96	0.96
Panel C: Tangibility						
Post×Treated	0.0195	0.0038*	0.0022	0.0627***	–0.0076***	–0.0051
	(0.0151)	(0.0022)	(0.0026)	(0.0103)	(0.0029)	(0.0032)
Obs.	286,269	980,046	991,310	256,779	851,176	866,343
Pseudo R-Squared	0.75	0.95	0.95	0.77	0.96	0.96
Panel D: Receivables/Sales						
Post×Treated	0.0517***	0.0008	0.0012	–0.0013	0.0036	0.0062
	(0.0089)	(0.0021)	(0.0024)	(0.0167)	(0.0036)	(0.0042)
Obs.	387,891	1,024,816	1,042,553	151,697	797,122	805,580
Pseudo R-Squared	0.75	0.95	0.95	0.77	0.96	0.96

D. Appendix: Banks' Use of Information

D.1 Robustness Tests: Dispersion

Table IA.XIII estimates the same specifications of Table 10 but without taking the logarithm of the interquartile range or the standard deviation.

Table IA.XIII. Robustness: Dispersion of Interest Rates and Probabilities of Default (levels)

This table reports estimates of the effect of eInvoicing on the dispersion of interest rates and banks' probability-of-default (PD) estimates at the province-quarter level. Panel A uses the dispersion of interest rates as the dependent variable; Panel B uses the dispersion of banks' PD estimates. Columns 1–3 measure dispersion using the interquartile range, while columns 4–6 measure dispersion using the standard deviation. Columns 1 and 4 restrict the sample to invoice firms; columns 2 and 5 restrict the sample to non-invoice firms; columns 3 and 6 use the full sample and estimate separate effects for invoice and non-invoice firms. For columns 1, 2, 4, and 5, we estimate:

$$y_{p,t} = \beta \cdot Post_t \times Treat_p + \alpha_p + \alpha_t + \Gamma \cdot X_{p,t} + \varepsilon_{p,t}.$$

For columns 3 and 6, we estimate the corresponding stacked specification:

$$y_{p,g,t} = \beta_1 \cdot Post_t \times Treat_p + \beta_2 \cdot Post_t \times Treat_p \times Invoice\ Firms_g + \alpha_{p,g} + \alpha_{g,t} + \Gamma_1 \cdot X_{p,t} + \Gamma_2 \cdot X_{p,t} \times Invoice\ Firms_g + \varepsilon_{p,g,t}.$$

In these equations, y is either the interquartile range or the standard deviation of interest rates or PDs within a province-quarter cell. In the stacked specification, g indexes whether the dispersion measure is computed among invoice firms or non-invoice firms, and $Invoice\ Firms_g$ is an indicator equal to one for the invoice-firms group. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for treated provinces. The vector $X_{p,t}$ includes log total credit, the number of firms, the corporate credit-market HHI, the share of ICO-guaranteed credit, the shares of credit to Agriculture, Mining, Manufacturing, Construction, Retail and Wholesale, Transport, and Hospitality, the Google Mobility index, and the total number of workers covered by the ERTE program, all constructed within the sample used to compute the dispersion measures. The row *Controls* indicates that all specifications include the vector $X_{p,t}$ (interacted with $Invoice\ Firms_g$ in the stacked specification). All variables are defined in Table IA.I. Heteroskedasticity-robust standard errors are reported in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

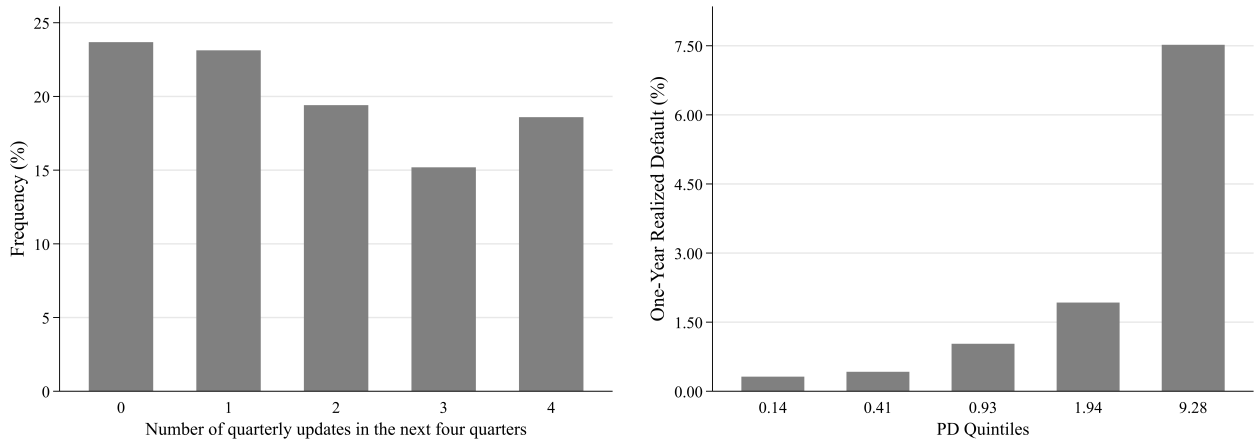
	Interquartile Range			Standard Deviation		
	Invoice	Non-Invoice	All	Invoice	Non-Invoice	All
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Interest Rates						
Post×Treated	9.35	–16.57	–16.57	14.86	–27.59	–27.59
	(12.49)	(13.51)	(13.51)	(11.42)	(21.96)	(21.96)
Post×Treated×Invoice Firms			25.92			42.45*
			(18.40)			(24.75)
Province FE + Quarter FE	Yes	Yes	No	Yes	Yes	No
Province×Invoice FE	No	No	Yes	No	No	Yes
Invoice×Quarter FE	No	No	Yes	No	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	96	96	192	96	96	192
R-Squared	0.81	0.85	0.86	0.73	0.81	0.89
Panel B: Probabilities of Default						
Post×Treated	30.58**	–22.38*	–22.38*	51.58	–70.88*	–70.88*
	(11.62)	(13.23)	(13.23)	(34.62)	(39.72)	(39.72)
Post×Treated×Invoice Firms			52.96***			122.45**
			(17.60)			(52.69)
Province FE + Quarter FE	Yes	Yes	No	Yes	Yes	No
Province×Invoice FE	No	No	Yes	No	No	Yes
Invoice×Quarter FE	No	No	Yes	No	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	96	96	192	96	96	192
R-Squared	0.74	0.73	0.74	0.78	0.66	0.76

D.2 Basic Facts about PDs

In this subsection, we validate that, in our sample, banks' internal estimates of borrowers' probability of default measure banks' private information.[45] We present two facts. First, PDs are updated frequently even after loan origination, suggesting that banks' information production plays an active role throughout the loan's lifespan. Figure IA.IVa shows the quarterly frequency of PD updates over a year. Banks change the PDs for approximately 76% of their outstanding loans over a year. Second, PDs are related to ex-post defaults. Figure IA.IVb shows that higher values of the PDs are positively correlated with an increased likelihood of ex-post default.

Figure IA.IV. Probabilities of Default (PDs): Updates and Validity

Panel (a) reports the quarterly frequency with which banks update the probabilities of default (PDs) of their outstanding loans over a year. Panel (b) plots ex-post loan defaults against the ex-ante PDs reported by IRB banks. The sample is the set of bank-firm-quarter observations with available PDs described in Section 5.2.



(a) Frequency of banks' PD updates

(b) Ex-post loan defaults vs ex-ante PDs

To show this more formally, we estimate the following regression at the loan level, following the same approach as Howes and Weitzner (2026):

$$Default_{l,t+4} = \beta \cdot \text{PD}_{l,t} + \Omega \cdot X_l + \delta_{b,t} + \gamma_{i,t} + \sigma_{b,p} + \varepsilon_l \tag{6}$$

where l, b, t, i and p index loan, bank, quarter, industry, and province, respectively. $Default_{l,t+4}$ is a dummy variable that equals one if loan l outstanding at t enters default or becomes doubtful within the subsequent four quarters. $\text{PD}_{l,t}$ denotes the one-year probability of default estimated by bank b at quarter t for loan l. X_l is a vector of firm- and loan-level characteristics, including firm size (measured as the logarithm of total assets), leverage (total debt to total assets), profitability (EBIT to total assets), asset tangibility (tangible assets to total assets), liquidity,

45. Howes and Weitzner (2026) and Casado and Martinez-Miera (2025) show this for the United States and Spain, respectively.

and the logarithm of one plus the firm's age. Loan-level contract controls include the logarithm of the loan credit volume in euros (including both drawn and undrawn amounts for credit lines), a binary indicator for whether the loan is secured with collateral, the proportion of the total credit volume granted to a firm by a specific bank within a given period, and loan type fixed effects. $\delta_{b,t}$ are bank×quarter fixed effects, $\gamma_{i,t}$ are industry×quarter fixed effects, and $\sigma_{b,p}$ are bank×province fixed effects. The coefficient of interest is β, which measures the expected change in realized default (in percentage points) per one-point increase in the loan's PD. Estimating Equation (6) with and without the loan interest rate as a control allows us to gauge whether the PD has predictive power beyond the credit risk information already embedded in the interest rate. As expected, the PD has a positive and highly significant correlation with the loan default, even after accounting for interest rates (see Table IA.XIV).

Table IA.XIV. Predictive Power of PDs

This table reports the coefficients of the loan-level regression specified in Equation (6), estimated by OLS. The dependent variable is a dummy variable equal to 1 if the loan enters default or becomes doubtful within the subsequent 4 quarters. Contract controls include the logarithm of the loan credit volume in euros (including both drawn and undrawn amounts for credit lines), a binary indicator for whether the loan is secured with real collateral, and the share of the total credit volume granted to a firm by a specific bank within a given period. Firm controls include leverage, liquidity, tangibility, profitability, log total assets, and log(1 + firm age). When we control for the interest rate, we also include an indicator for the loan's interest-rate structure, which equals 1 for fixed-rate loans and 0 otherwise. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province × quarter level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

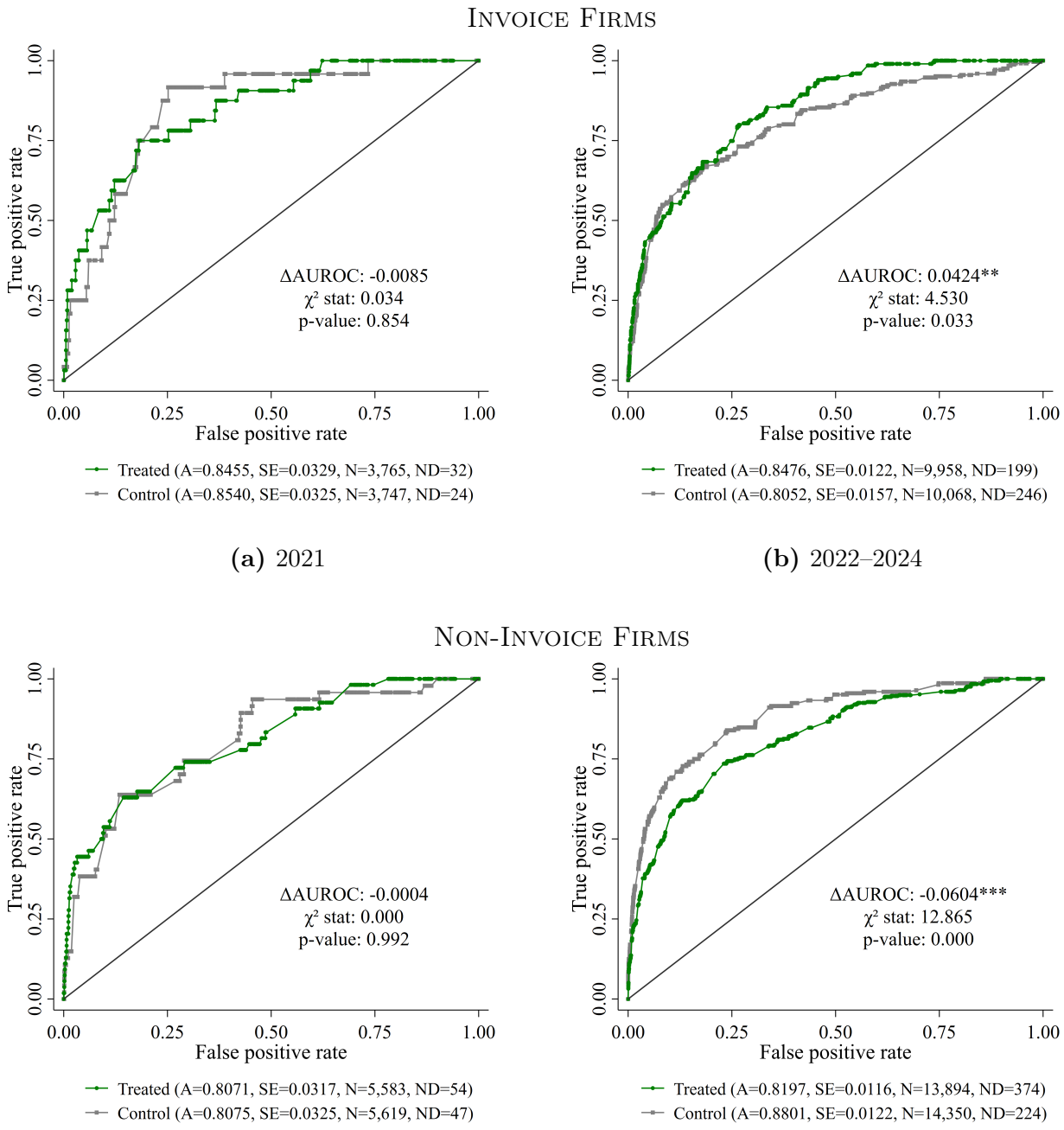
	Loan Default			
	(1)	(2)	(3)	(4)
PD	0.5087***	0.5072***	0.4732***	0.4879***
	(0.0165)	(0.0165)	(0.0163)	(0.0155)
Interest Rate				0.0882***
				(0.0143)
Fixed Interest Rate Indicator				0.0060
				(0.0473)
Industry×Quarter FE	Yes	Yes	Yes	Yes
Bank×Province FE	Yes	Yes	Yes	Yes
Bank×Quarter FE	Yes	Yes	Yes	Yes
Loan Type FE	No	Yes	Yes	Yes
Contract Controls	No	Yes	Yes	Yes
Firm Controls	No	No	Yes	Yes
Obs.	826,425	826,425	826,425	541,939
R-Squared	0.06	0.07	0.07	0.08

D.3 Robustness Tests: AUROCs

D.3.1 Matched/Reweighted Samples

Figure IA.V. Robustness: Differences in Predictive Quality of PDs before and after eInvoicing, Matched/Reweighted Sample

This figure replicates the AUROC-based analysis in Figure 3 on a matched/reweighted sample in which treated and control firms are balanced on observable characteristics. The top row reports AUROCs for invoice firms, and the bottom row for non-invoice firms. The implied difference-in-differences is 0.051 for invoice firms and −0.060 for non-invoice firms. The larger the area under the curve, the higher the accuracy of the estimated PDs. Legend, test statistics, and significance levels as in Figure 2.



(a) 2021 **(b)** 2022–2024

(c) 2021 **(d)** 2022–2024

Table IA.XV. Descriptive Statistics for the AUROC Sample: Invoice vs. Non-Invoice Firms, Treated vs. Control (Pre- and Post-Matching)

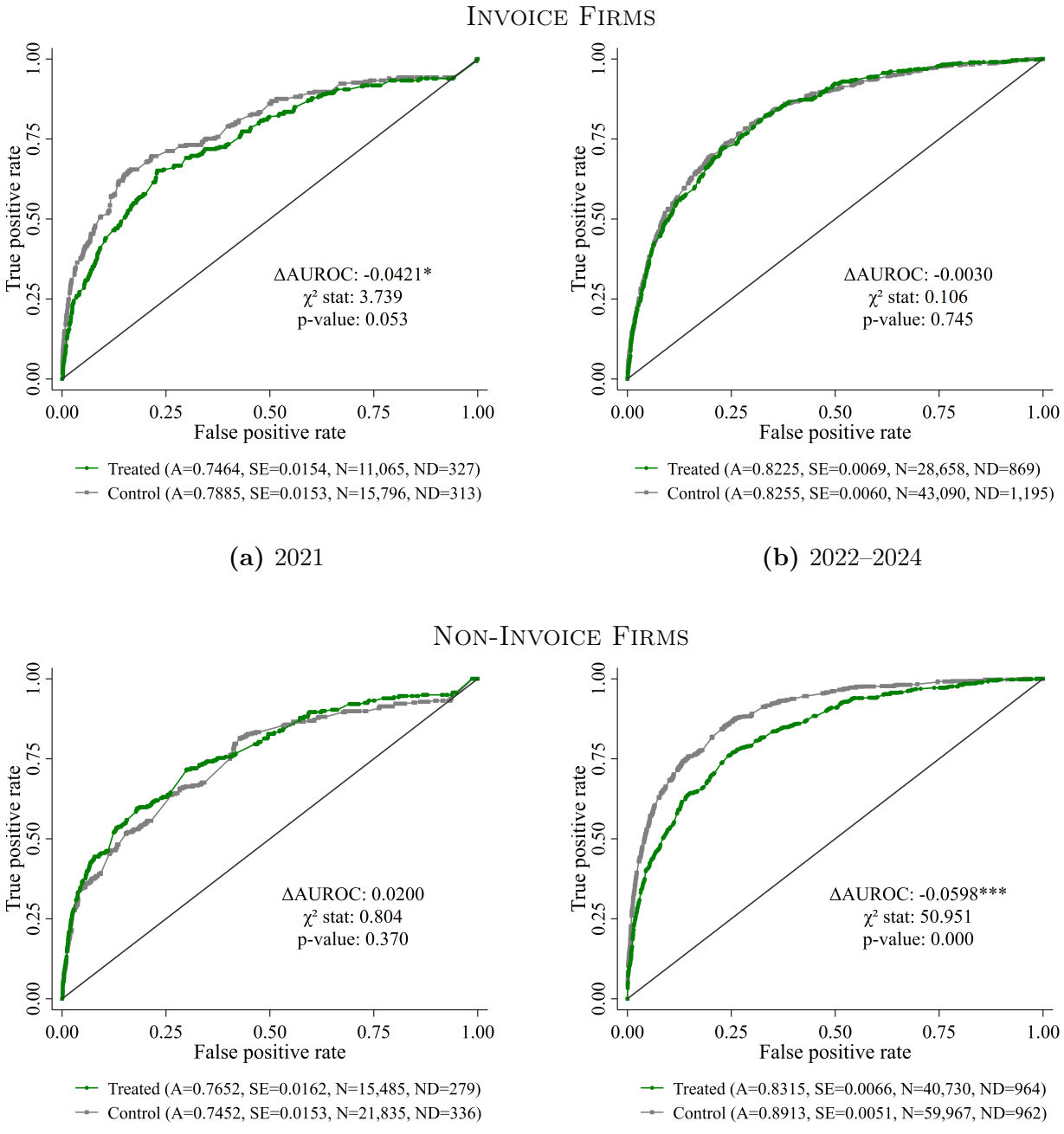
This table presents descriptive statistics for the AUROC sample. All variables are measured as of the end of 2021, the pre-reform period, and are defined in Table IA.I. Panel A reports pre-matching statistics, while Panel B reports post-matching statistics. Within each panel, observations are further split into Invoice firms and Non-Invoice firms, and into control and treated groups. Columns (1) and (2) report the mean for the treated and control groups, respectively. Column (3) reports the difference between the treated and control groups, and Column (4) reports the p-value of a two-sided test for the equality of means across the treated and control groups. Standard deviations appear in parentheses, except for N, for which the relative share appears in parentheses. p-values are reported to two decimal digits. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Treated	Control	Diff.	P-value
	(1)	(2)	(3)	(4)
Panel A: Pre-Matching				
Invoice Firms				
N	2,719 (40.70%)	3,962 (59.30%)		
Total credit, thousand €	335.15 (560.50)	396.20 (666.66)	–61.05	0.15
Median PD	2.81 (7.05)	2.35 (5.84)	0.45	0.03**
Size	7.54 (1.35)	7.61 (1.38)	–0.07	0.63
Leverage	0.38 (0.23)	0.38 (0.21)	0.00	0.68
Liquidity	0.73 (0.22)	0.68 (0.22)	0.05	0.08*
Tangibility	0.19 (0.20)	0.25 (0.20)	–0.05	0.03**
Profitability	0.02 (0.13)	0.03 (0.09)	–0.01	0.35
Age	21.49 (12.09)	23.01 (12.66)	–1.52	0.22
Share of ICO-guaranteed credit	0.49 (0.40)	0.48 (0.40)	0.01	0.66
Non-Invoice Firms				
N	3,859 (41.03%)	5,547 (58.97%)		
Total credit, thousand €	201.17 (536.34)	183.24 (460.92)	17.93	0.48
Median PD	2.11 (6.00)	1.96 (5.76)	0.15	0.47
Size	6.47 (1.53)	6.33 (1.46)	0.14	0.32
Leverage	0.35 (0.31)	0.36 (0.29)	–0.01	0.83
Liquidity	0.63 (0.31)	0.59 (0.30)	0.05	0.38
Tangibility	0.23 (0.27)	0.30 (0.27)	–0.07	0.02**
Profitability	0.03 (0.18)	0.03 (0.16)	–0.01	0.28
Age	17.23 (11.38)	17.77 (11.62)	–0.53	0.57
Share of ICO-guaranteed credit	0.44 (0.44)	0.47 (0.44)	–0.03	0.45
Panel B: Post-Matching				
Invoice Firms				
N	997 (50.00%)	997 (50.00%)		
Total credit, thousand €	358.31 (560.87)	383.35 (597.91)	–25.04	0.56
Median PD	1.77 (3.35)	1.87 (3.77)	–0.09	0.68
Size	7.70 (1.27)	7.65 (1.20)	0.04	0.80
Leverage	0.37 (0.21)	0.36 (0.20)	0.01	0.52
Liquidity	0.75 (0.21)	0.75 (0.19)	0.00	0.92
Tangibility	0.19 (0.19)	0.19 (0.17)	0.00	0.94
Profitability	0.03 (0.08)	0.03 (0.07)	0.00	0.86
Age	22.31 (12.09)	22.98 (11.83)	–0.67	0.57
Share of ICO-guaranteed credit	0.46 (0.40)	0.47 (0.40)	–0.01	0.84
Non-Invoice Firms				
N	1,487 (50.00%)	1,487 (50.00%)		
Total credit, thousand €	145.80 (379.37)	159.92 (388.96)	–14.12	0.72
Median PD	1.60 (2.74)	1.72 (4.17)	–0.12	0.63
Size	6.04 (1.39)	6.05 (1.40)	–0.01	0.97
Leverage	0.38 (0.31)	0.36 (0.31)	0.01	0.70
Liquidity	0.64 (0.32)	0.64 (0.30)	0.00	0.98
Tangibility	0.26 (0.28)	0.26 (0.26)	0.00	0.94
Profitability	0.03 (0.16)	0.03 (0.16)	0.00	0.91
Age	16.16 (10.91)	16.93 (10.90)	–0.77	0.54
Share of ICO-guaranteed credit	0.48 (0.45)	0.50 (0.45)	–0.02	0.81

D.3.2 Stayers Samples

Figure IA.VI. Robustness: Differences in Predictive Quality of PDs before and after eInvoicing, Stayers Only

This figure replicates the AUROC-based analysis in Figure 3, restricting the sample to *stayers*, i.e., bank-firm relationships that are present in the data both before and after the reform. The top row reports AUROCs for invoice firms, and the bottom row for non-invoice firms. The implied difference-in-differences is 0.039 for invoice firms and −0.080 for non-invoice firms. The larger the area under the curve, the higher the accuracy of the estimated PDs. Legend, test statistics, and significance levels as in Figure 2.



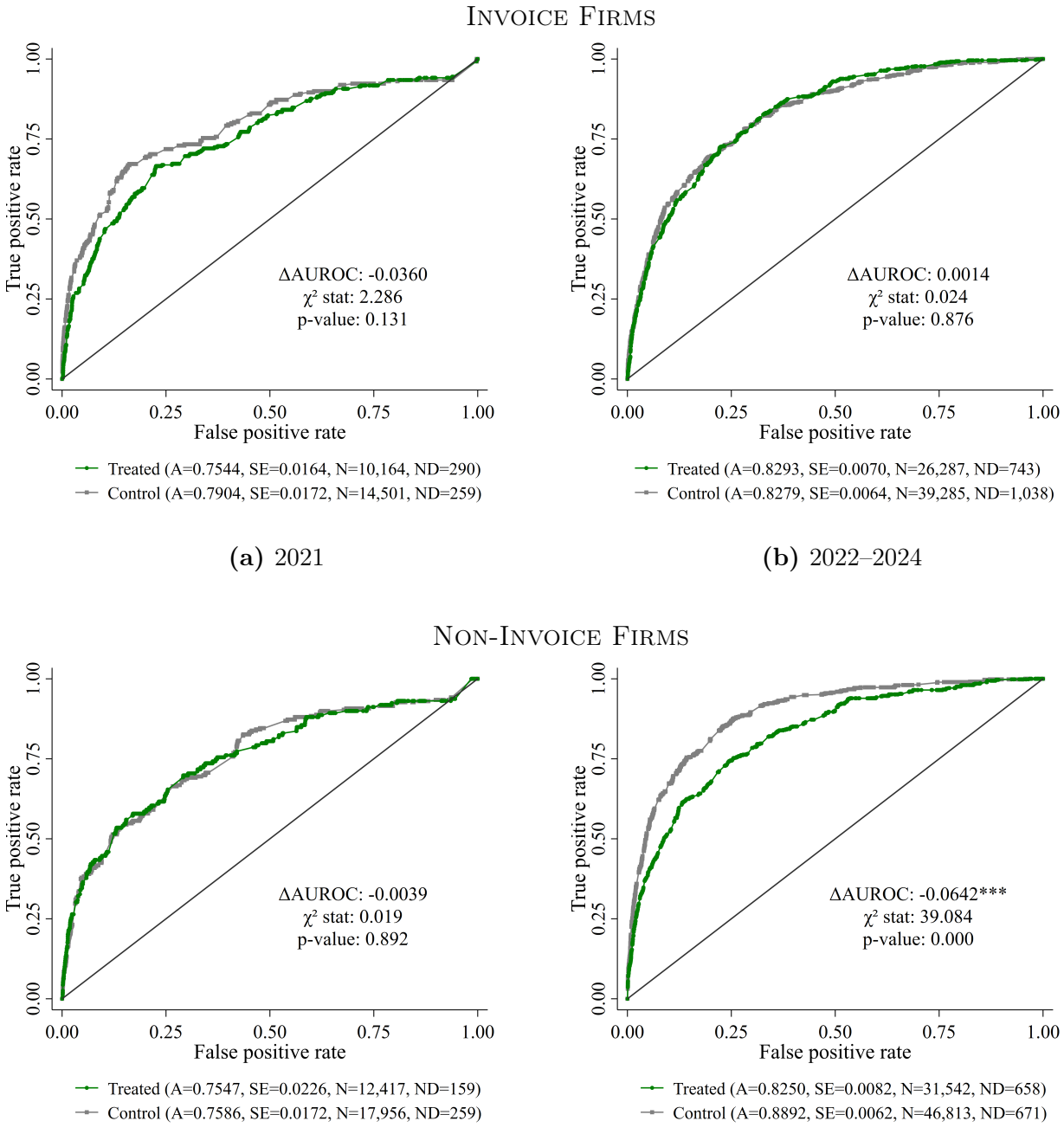
(a) 2021 **(b)** 2022–2024

(c) 2021 **(d)** 2022–2024

D.3.3 Same-Cohort Samples

Figure IA.VII. Robustness: Differences in Predictive Quality of PDs before and after eInvoicing, Relationships Starting in 2019 or Earlier

This figure replicates the AUROC-based analysis in Figure 3 for bank-firm relationships that begin in 2019 or earlier. The top row reports AUROCs for invoice firms, and the bottom row for non-invoice firms. The implied difference-in-differences is 0.037 for invoice firms and −0.060 for non-invoice firms. The larger the area under the curve, the higher the accuracy of the estimated PDs. Legend, test statistics, and significance levels as in Figure 2.

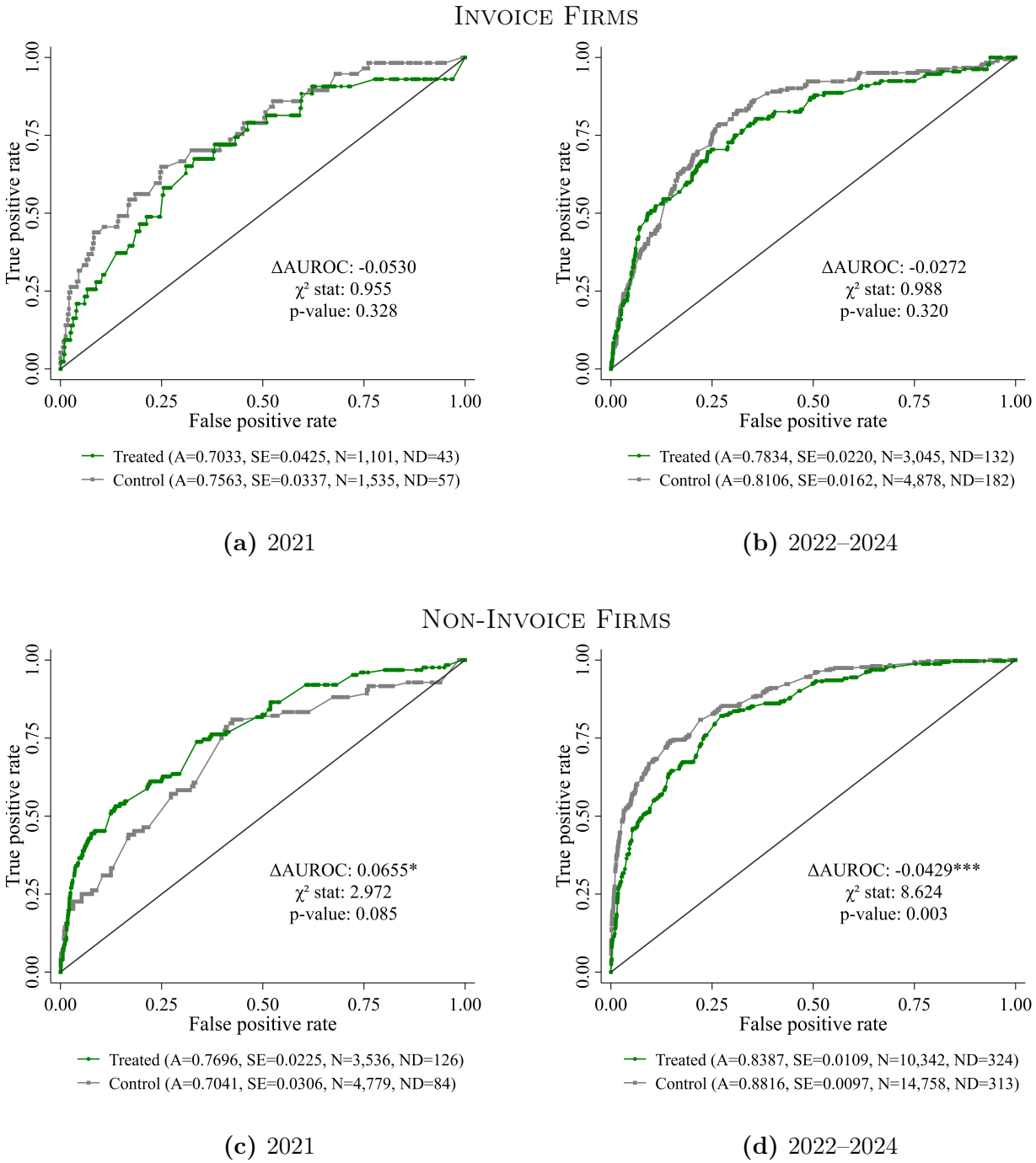


(a) 2021 **(b)** 2022–2024

(c) 2021 **(d)** 2022–2024

Figure IA.VIII. Robustness: Differences in Predictive Quality of PDs before and after eInvoicing, Relationships Starting in 2020–2021

This figure replicates the AUROC-based analysis in Figure 3 for bank-firm relationships that begin in 2020-2021. The top row reports AUROCs for invoice firms, and the bottom row for non-invoice firms. The implied difference-in-differences is 0.026 for invoice firms and −0.108 for non-invoice firms. The larger the area under the curve, the higher the accuracy of the estimated PDs. Legend, test statistics, and significance levels as in Figure 2.



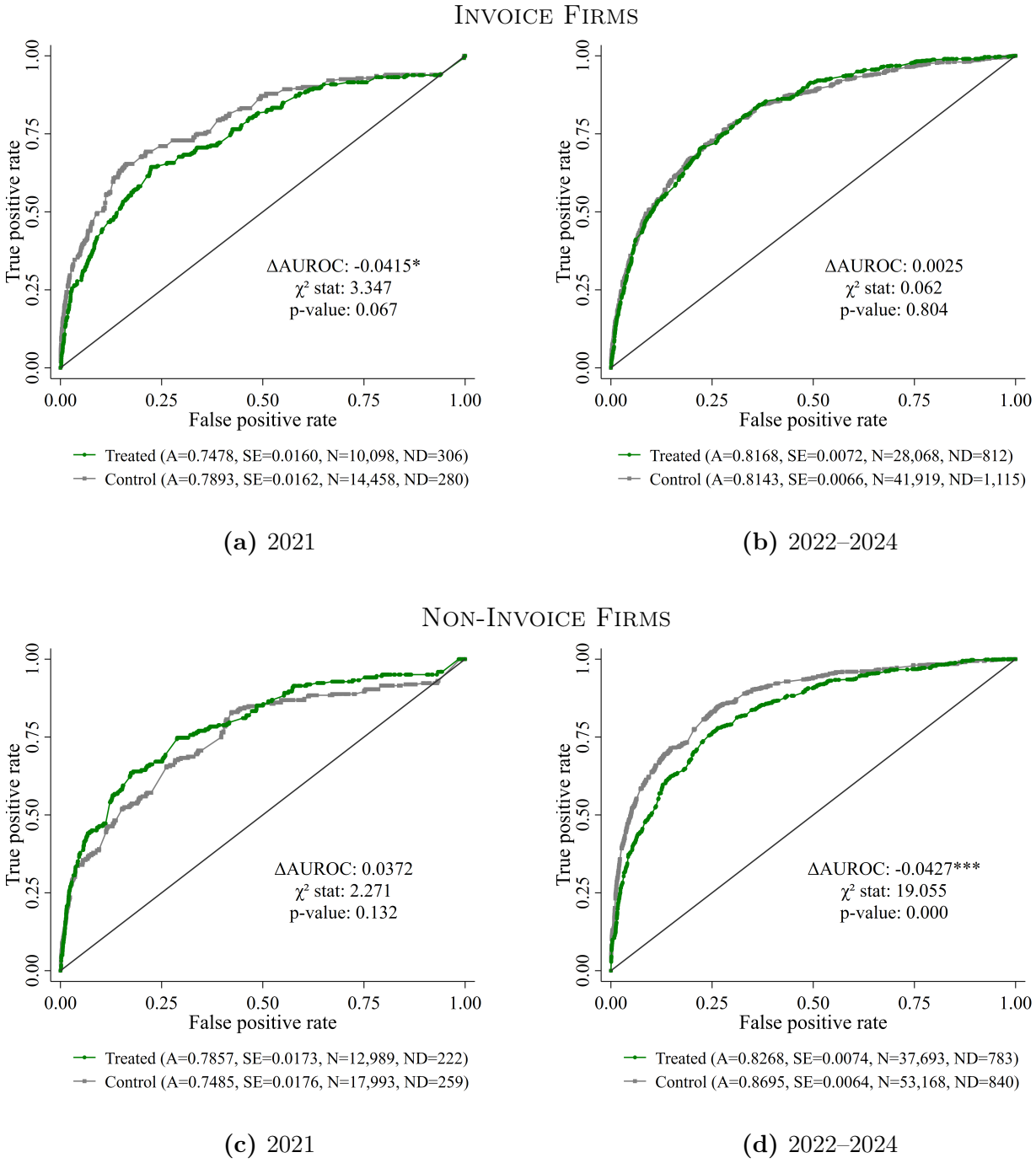
(a) 2021 **(b)** 2022–2024

(c) 2021 **(d)** 2022–2024

D.3.4 Samples without ICO-guaranteed or Moratoria-related Loans

Figure IA.IX. Robustness: Differences in Predictive Quality of PDs before and after eInvoicing, Excluding ICO-Guaranteed and Moratoria Loans

This figure replicates the AUROC-based analysis in Figure 3 after dropping all bank-firm-quarter observations in which credit consists solely of ICO-guaranteed loans or loans affected by the default moratoria. The top row reports AUROCs for invoice firms, and the bottom row for non-invoice firms. The implied difference-in-differences is 0.044 for invoice firms and −0.080 for non-invoice firms. The larger the area under the curve, the higher the accuracy of the estimated PDs. Legend, test statistics, and significance levels as in Figure 2.



(a) 2021 **(b)** 2022–2024

(c) 2021 **(d)** 2022–2024

D.4 Credit Analysis with AUROC Matching Procedure

In this section, we repeat our main credit analysis on a sample of firms obtained using the same matching procedure as in the AUROC-based analysis. Specifically, we match bank-firm-quarter observations across treated and control groups, applying the procedure described in Section 5.2.4 and employing the same set of variables, except for banks' IRB PD estimates for the firm. The results are reported in Table IA.XVI and support a credit reallocation toward invoice firms, as in our baseline analysis.

Table IA.XVI. Robustness: Matched Samples

This table reports estimates of β from regressions estimated separately for invoice firms and non-invoice firms over matched samples from January 2021 to December 2024, using Poisson Pseudo-Maximum Likelihood. The unit of observation is a bank-firm-month, and we estimate:

$$\mathbb{E}[Y_{f,b,t} \mid \cdot] = \exp\left\{\alpha_{i,s,t} + \alpha_{f,b} + \alpha_{b,t} + \beta \cdot Post_t \times Treat_p + \Gamma_f \cdot X_{f,t-1} + \Gamma_p \cdot X_{p,i,t}\right\}.$$

$Y_{f,b,t}$ is the amount of credit extended by bank b to firm f in month t. For invoice firms, the dependent variable is invoice-based credit in column 1, non-invoice credit in column 2, and total credit in column 3. For non-invoice firms, the dependent variable is total credit in column 4. We cannot estimate specifications with invoice-based credit as the dependent variable for non-invoice firms because, by construction, these firms have no positive invoice-based credit in the pre-reform period. $Post_t$ is an indicator equal to one from January 2022 onward. $Treat_p$ is an indicator equal to one for firms headquartered in treated provinces. The specification includes industry×size-tercile×month, firm×bank, and bank×month fixed effects. $X_{f,t-1}$ denotes the vector of previous-year firm controls, which includes firm size, leverage, liquidity, tangibility, profitability, age, the share of bank-firm credit guaranteed by ICO, and an indicator for whether the firm is in default with that bank. $X_{p,i,t}$ denotes Covid controls, which include the number of employees covered by the ERTE temporary employment adjustment scheme in the firm's province-industry-month cell and the Google Mobility index at the province-month level. The row *Controls* indicates that all specifications include firm and Covid controls. All variables are defined in Table IA.I. Standard errors (in parentheses) are clustered at the province×month level. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Invoice Firms			**Non-Invoice Firms**
	Invoice	Non-Invoice	Total	Total
	(1)	(2)	(3)	(4)
Post×Treated	0.0588***	0.0013	0.0100**	–0.0011
	(0.0108)	(0.0045)	(0.0046)	(0.0026)
Industry×Size×Month FE	Yes	Yes	Yes	Yes
Firm×Bank FE	Yes	Yes	Yes	Yes
Bank×Month FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Obs.	136,148	182,136	187,003	311,599
Pseudo R-Squared	0.75	0.93	0.93	0.97